

BODY & ACCESSORIES

09

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09-00 OUTLINE

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BODY AND ACCESSORIES ABBREVIATIONS

dcf09000000t01

4W-ABS	4-Wheel Antilock Brake System
ABS	Antilock Brake System
ACC	Accessories
ALC	Auto Level Control
CM	Control Module
GND	Ground
HI	High
HU	Hydraulic Unit
IG	Ignition
INT	Intermittent
LCD	Liquid Crystal Display
LED	Light Emitting Diode
LH	Left Hand
LO	Low
M	Motor
OFF	Switch Off
ON	Switch On
PCM	Powertrain Control Module
RFW	Remote Free Wheel
RH	Right Hand
SW	Switch
TNS	Tail Number Side Lights

BODY AND ACCESSORIES FEATURES

dcf09000000t02

Improved marketability	• Power window system adopted • Power door lock system adopted • Keyless entry system adopted
Improved security	• Immobilizer system adopted • Theft-deterrent system adopted
Improved convenience	• A center panel unit, composed of the installed audio unit and the audio switches built into the center panel, has been adopted

09

09-10 BODY PANELS

BODY PANEL OUTLINE 09-10-1

BODY PANEL STRUCTURAL VIEW 09-10-1

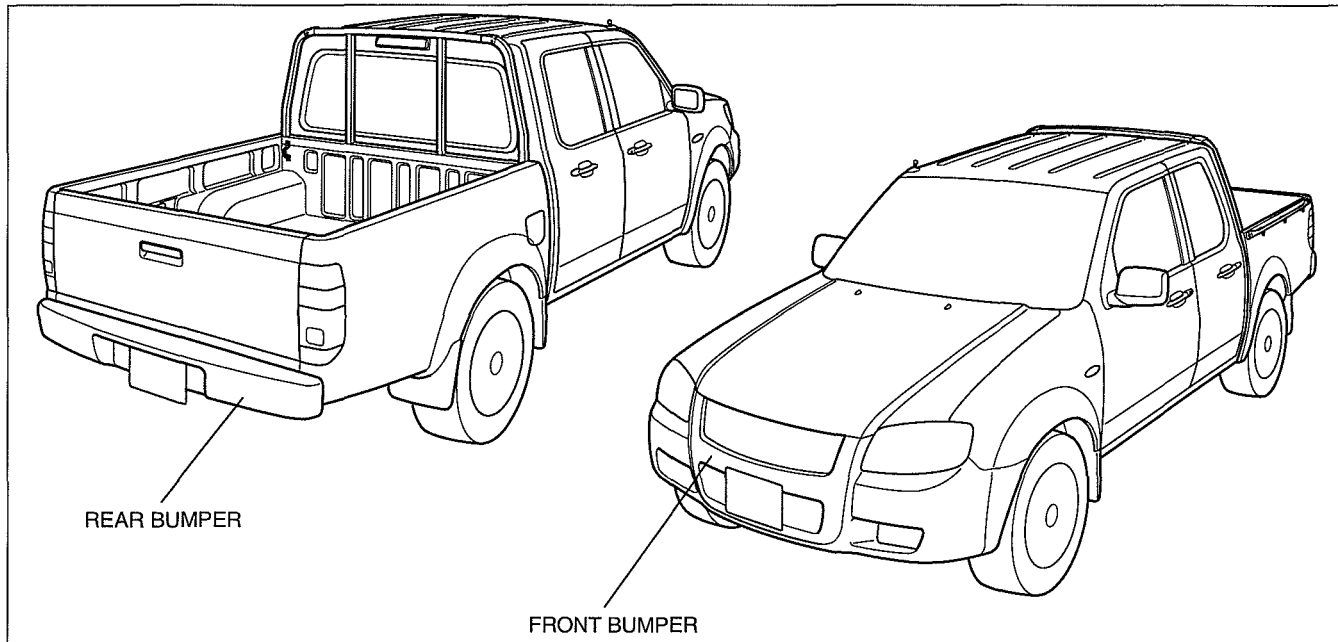
BODY PANEL OUTLINE

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- Front and Rear Bumper has been changed.

BODY PANEL STRUCTURAL VIEW

dcf091056100t02



DBG910ZTB001

09-12 GLASS/WINDOWS/MIRRORS

REAR WINDOW DEFROSTER		POWER WINDOW SYSTEM	
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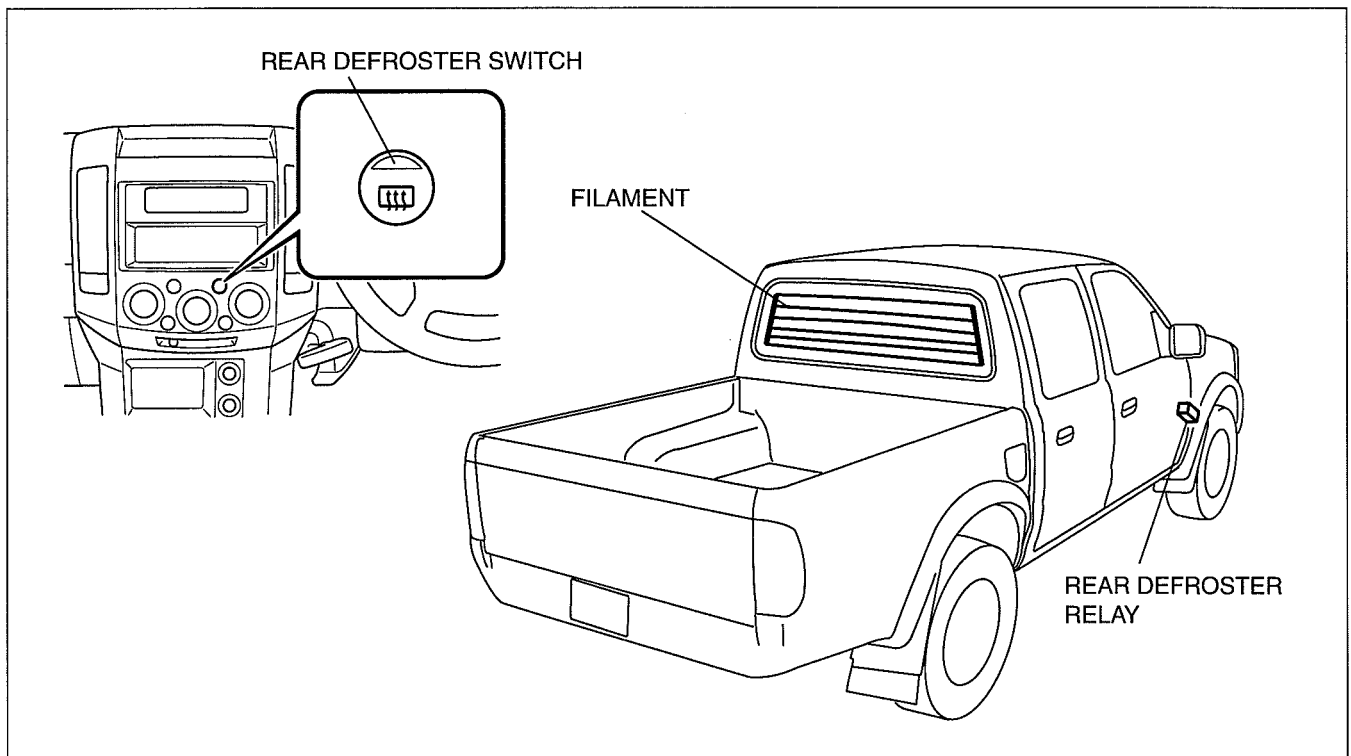
REAR WINDOW DEFROSTER OUTLINE

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- Rear window defroster system has been adopted.

REAR WINDOW DEFROSTER STRUCTURAL VIEW

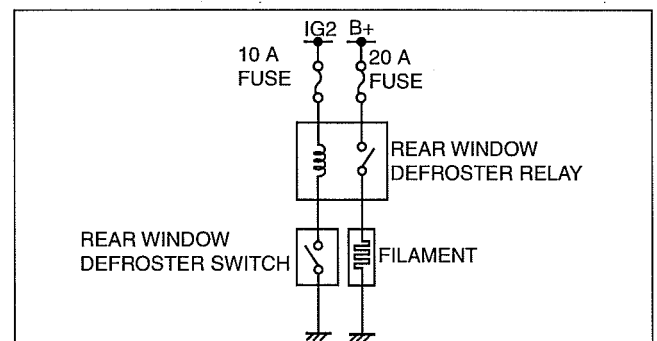
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DBR912ZT2001

REAR WINDOW DEFROSTER SYSTEM WIRING DIAGRAM

dcf091263000t03



DBG912ZTB451

GLASS/WINDOWS/MIRRORS

REAR WINDOW DEFROSTER SYSTEM OPERATION

dcf091263000t04

- When the rear window defroster switch is pressed, the rear window defroster turns on.
- Fogging is cleared from the rear window and outer mirror glass by heating of the filament.
- If the rear window defroster switch is pressed again with the rear window defroster on, defrosting will stop. Defroster operation also stops if the engine switch is turned to LOCK while the system is operating

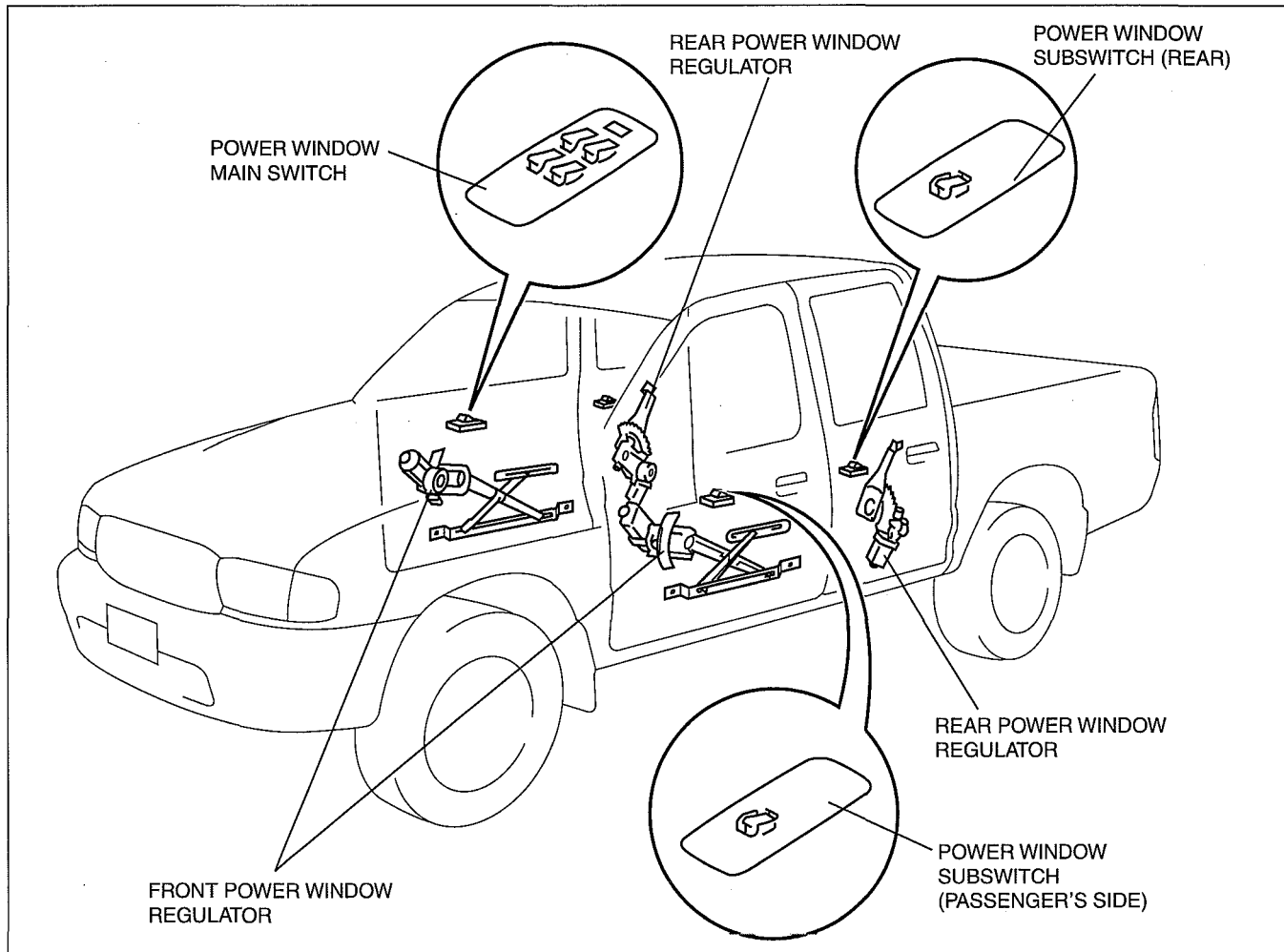
POWER WINDOW SYSTEM OUTLINE

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- The following functions have been adopted for all windows.
 - Manual open/close function
 - Auto-open function
- A power-cut function that permits disabling the operation of all window switches from the driver's seat has been adopted.

POWER WINDOW SYSTEM STRUCTURAL VIEW

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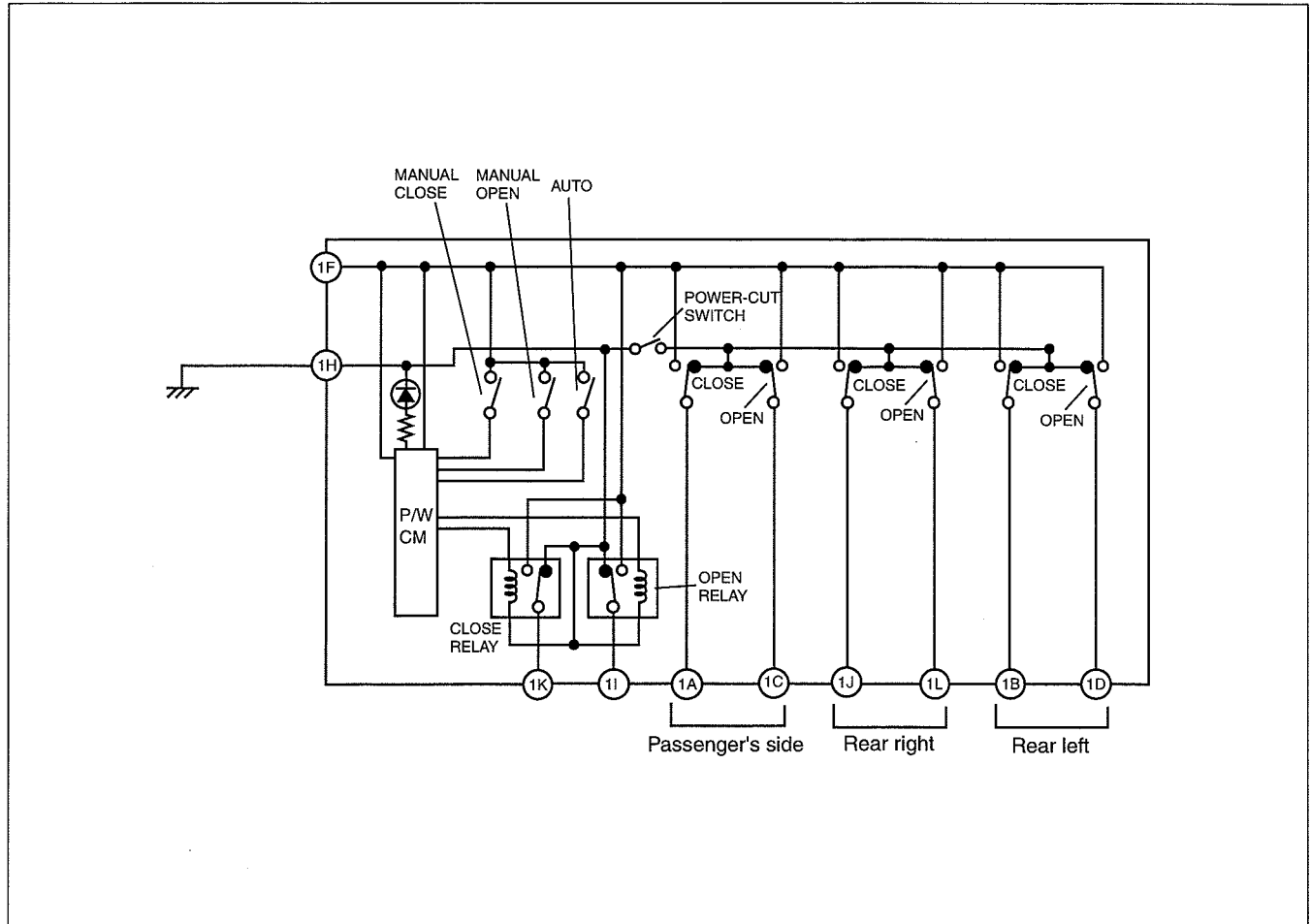


DBR912ZT2002

GLASS/WINDOWS/MIRRORS

POWER WINDOW SYSTEM WIRING DIAGRAM

dcf09125800t03



DCF912ZT2001

POWER WINDOW SYSTEM OPERATION

dcf09125800t06

Manual Open/Close

- The window opens/closes according to the down (push) or up (pull) operation of the power window main switches and the power window subswitches.

Auto-Open

- The window automatically moves to a fully-opened position when the power window main switches is operated to fully open positions, either down (push).

POWER OUTER MIRROR OUTLINE

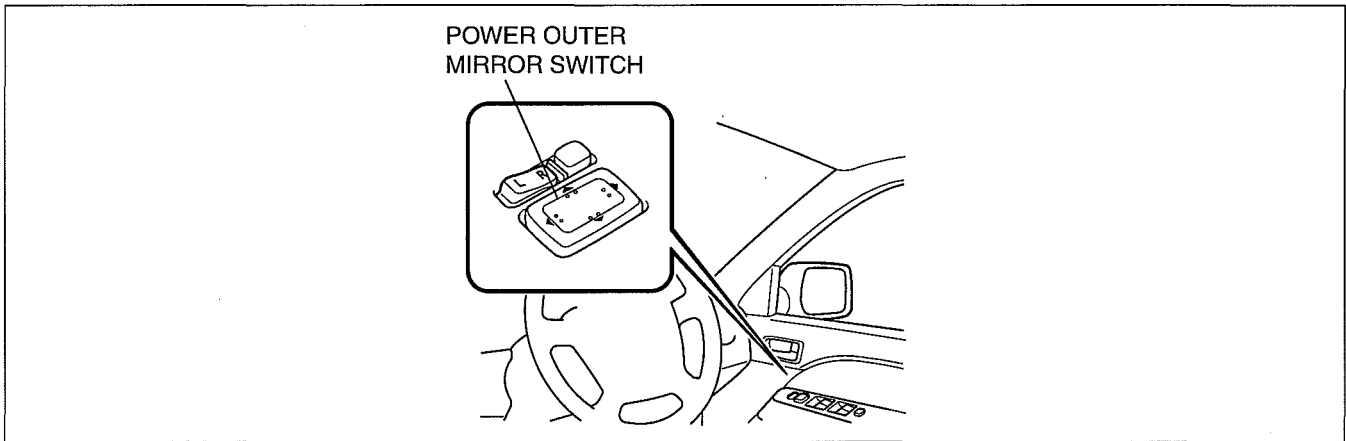
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Improved convenience	Power outer mirror (mirror glass adjusting function) adopted
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GLASS/WINDOWS/MIRRORS

POWER OUTER MIRROR STRUCTURAL VIEW

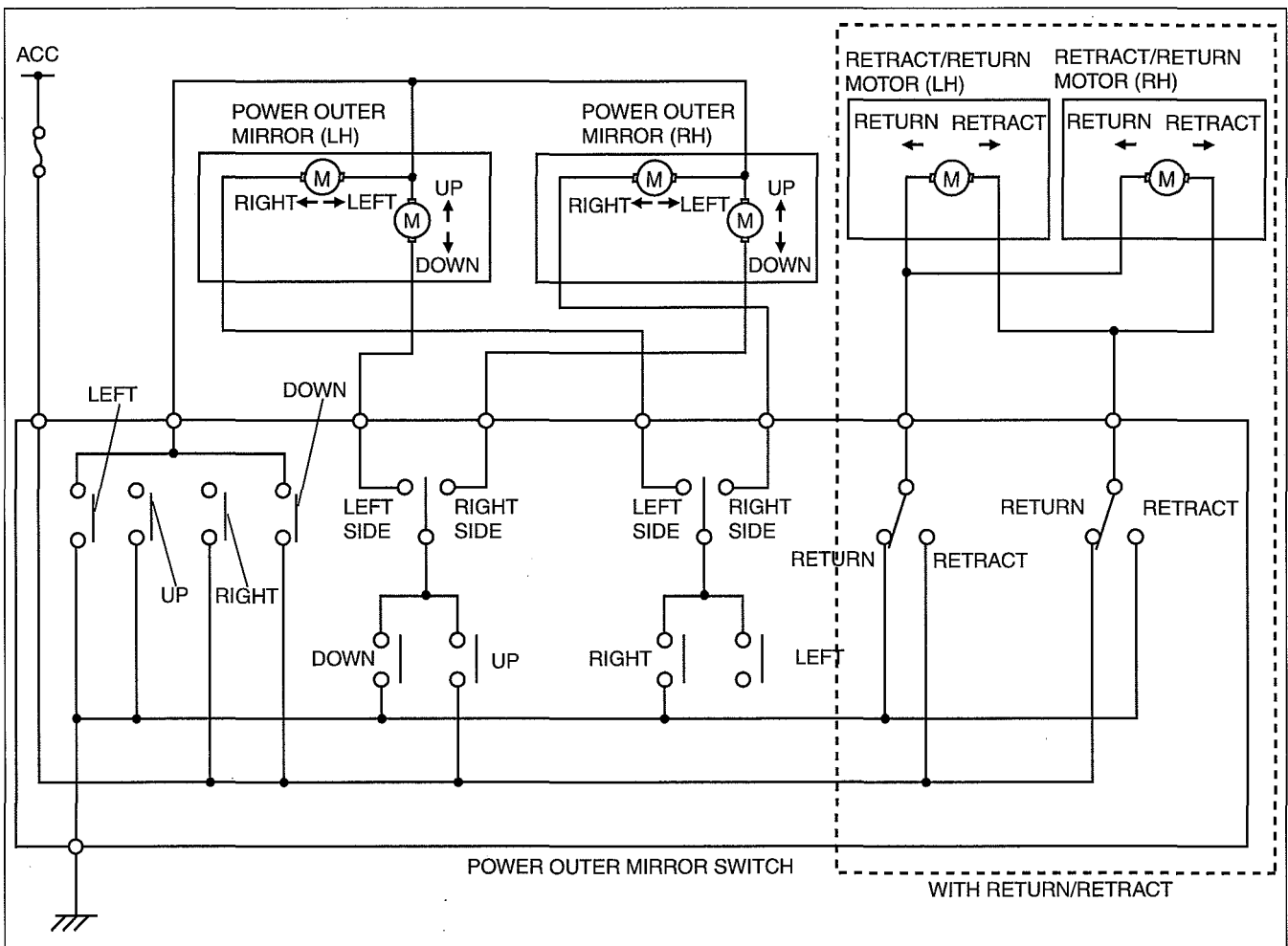
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DBR912ZT2004

POWER OUTER MIRROR SYSTEM WIRING DIAGRAM

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DBR912ZT2005

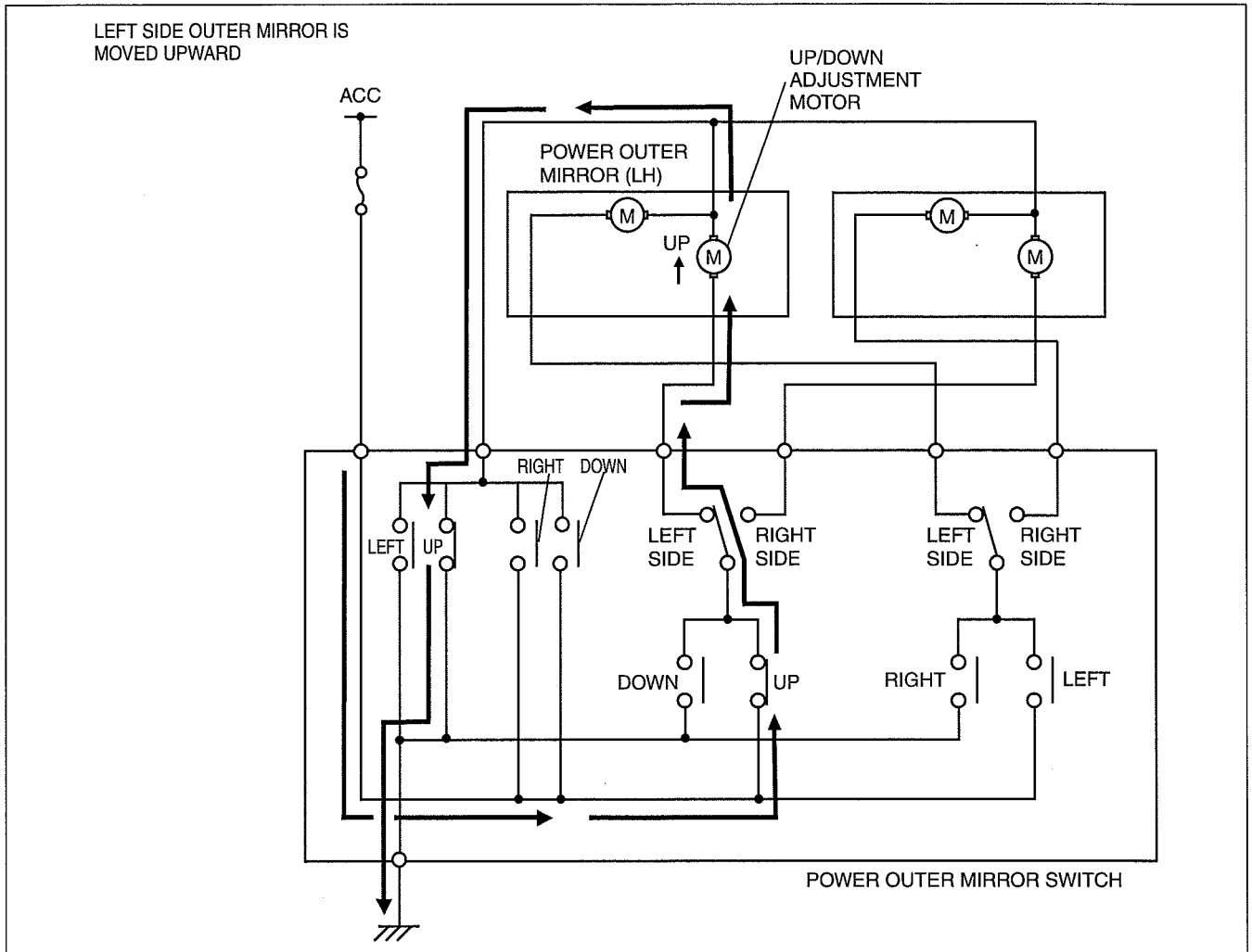
GLASS/WINDOWS/MIRRORS

POWER OUTER MIRROR OPERATION

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Mirror Glass Adjustment

- The left/right selection switch establishes left or right side outer mirror circuit and current is supplied in either one of the four directions according to the position of the mirror glass adjustment switch. Due to this, the motor rotates either up or down, left or right.



E5U912ZS5007

SEATS

09-13 SEATS

SEATS OUTLINE..... 09-13-1
SEATS SPECIFICATION..... 09-13-1

SEATS STRUCTURAL VIEW.....09-13-2

SEATS OUTLINE

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FEATURES

Improved marketability	• Two types of have been adopted for the driver and passenger seats. — Separate type — Bench type
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SEATS SPECIFICATION

dcf091357000t02

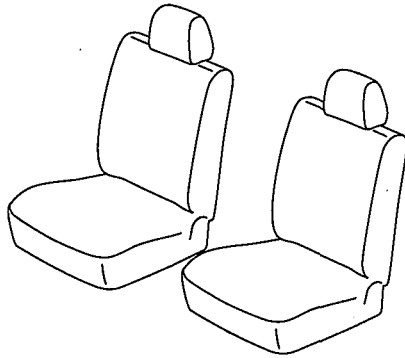
Body type	Item	Function
Stretch cab	Seat	Recliner
		Slide
		Side air bag module (See 08-10-5 SIDE AIR BAG MODULE FUNCTION.)
Regular cab	Seat	Recliner
		Slide
		Armrest
Double cab	Front seat	Recliner
		Slide
		Side air bag module (See 08-10-5 SIDE AIR BAG MODULE FUNCTION.)
	Rear seat	Armrest

SEATS

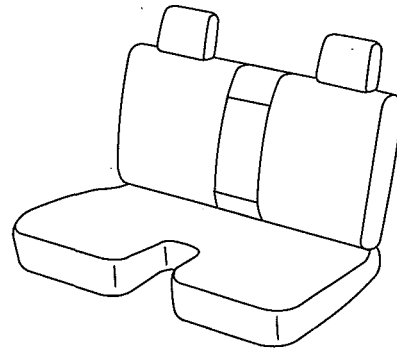
SEATS STRUCTURAL VIEW

dcf091357000t03

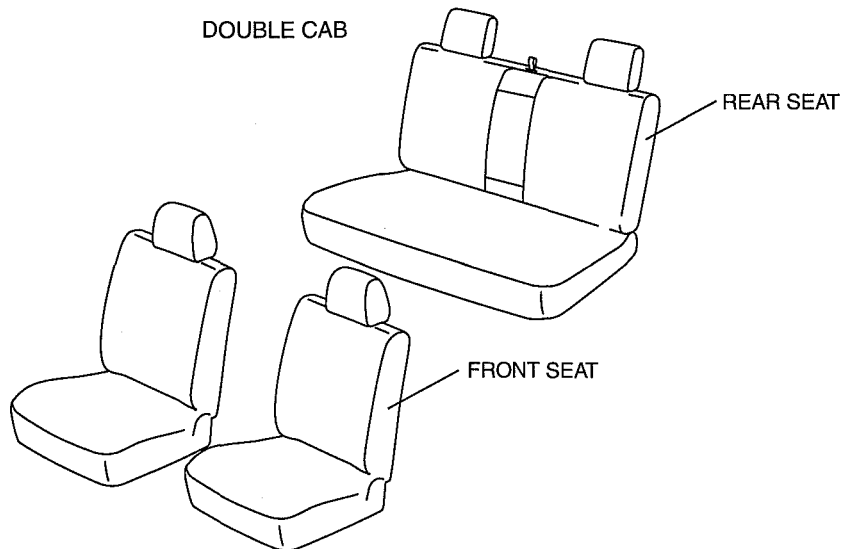
STRETCH CAB
(WITH REAR ACCESS SYSTEM)



REGULAR CAB



DOUBLE CAB



DCF913ZTB003

09-14 SECURITY AND LOCKS

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KEYLESS ENTRY SYSTEM	
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SECURITY AND LOCKS OUTLINE

dcf091400001t01

Improved marketability	• Power door lock system adopted • Keyless entry system adopted
Improved security	• Theft-deterrent system adopted • Immobilizer system adopted

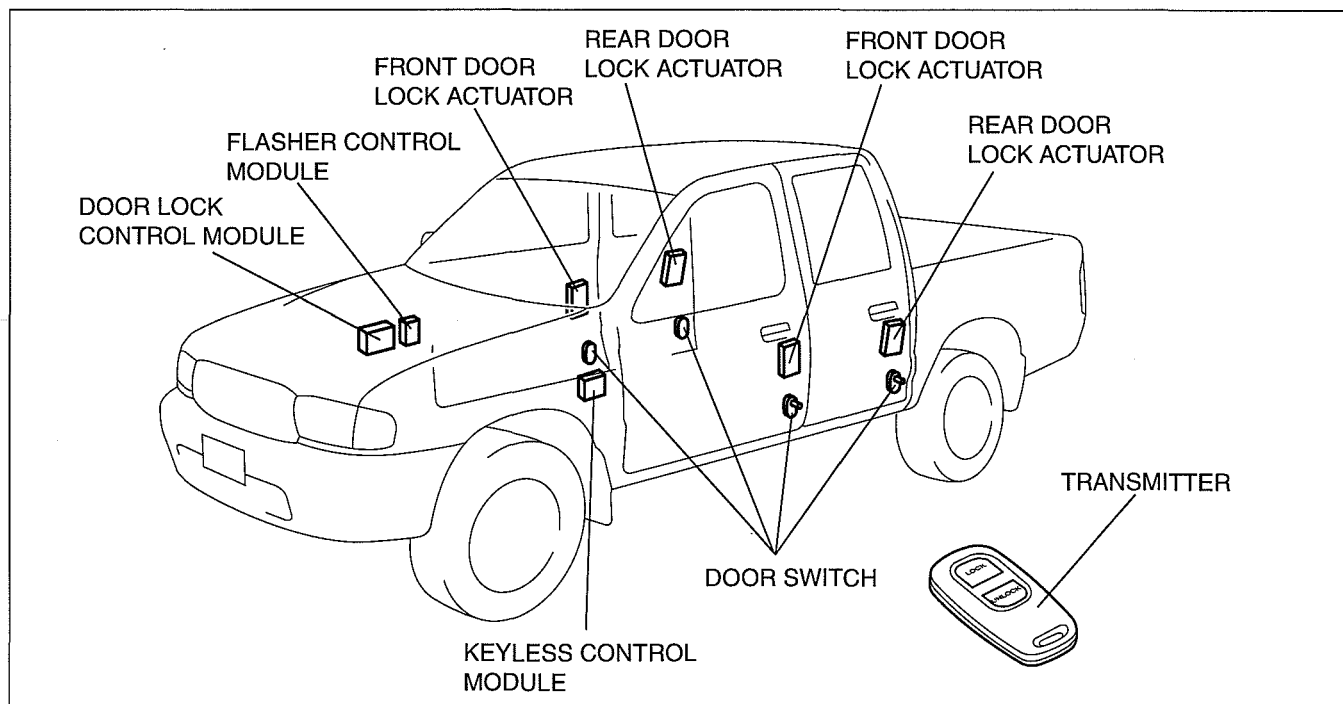
KEYLESS ENTRY SYSTEM OUTLINE

dcf091466000t01

- The doors also can be locked/unlocked by operating the key or transmitter.
- The answer-back function has been adopted where the hazard warning lights flash and a buzzer sounds to confirm that the doors are locked/unlocked.
- A rolling code type transmitter has been adopted to prevent theft by radiowave interception.
- To prevent improper operation while the vehicle is moving, the doors cannot be locked/unlocked by operating the transmitter if the ignition key is not in the LOCK position.

KEYLESS ENTRY SYSTEM STRUCTURAL VIEW

dcf091466000t02

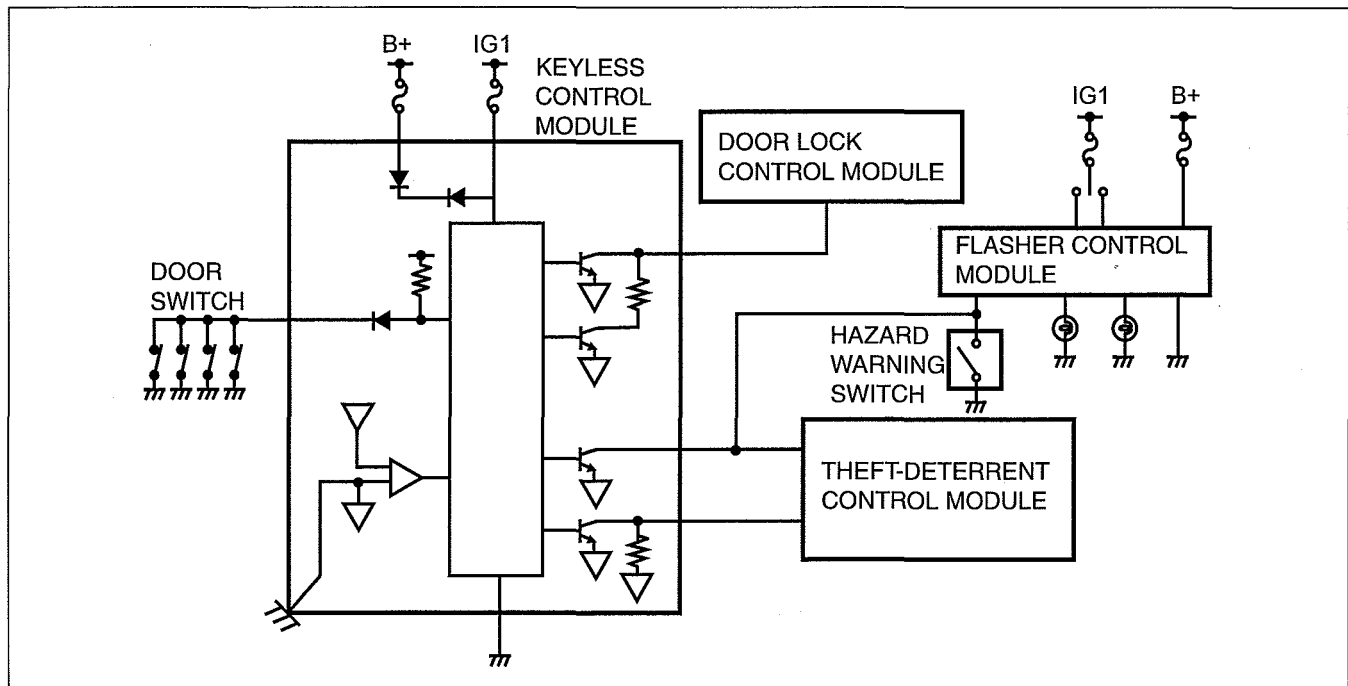


DFC914ZWZ001

SECURITY AND LOCKS

KEYLESS ENTRY SYSTEM WIRING DIAGRAM

dcf091466000t03



DBR914ZW8002

KEYLESS ENTRY SYSTEM OPERATION

dcf091466000t04

- When the transmitter LOCK/UNLOCK button is pushed, all doors lock/unlock.
- In response to transmitter operation (lock/unlock), the hazard warning lights flash to enable visual verification of operation.
 - When the transmitter LOCK button is pushed, the hazard warning lights flash once.
 - When the transmitter UNLOCK button is pushed, the hazard warning lights flash twice.
- When any door is not opened within 30 s after the transmitter UNLOCK button is pressed, the UNLOCK signal is cancelled and the doors are locked.

THEFT-DETERRENT SYSTEM OUTLINE

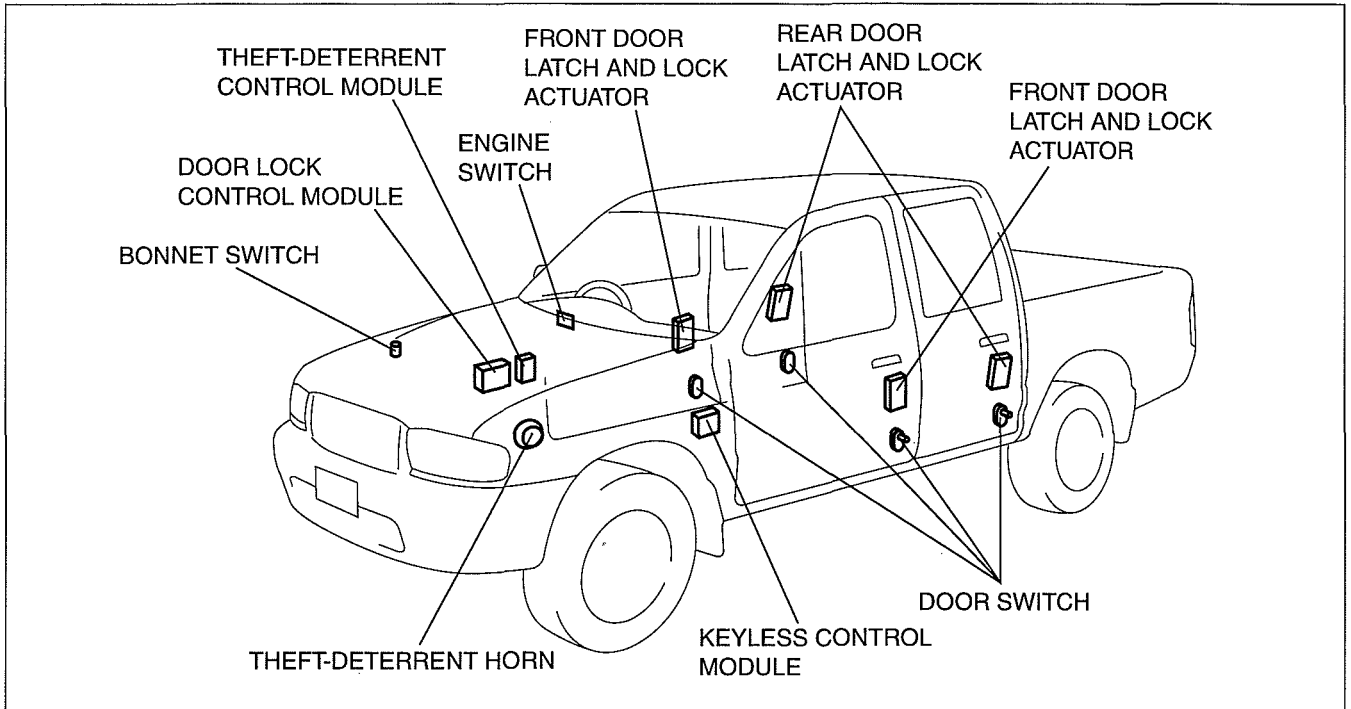
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- The theft-deterrent system includes sound and light alarms that activate when the bonnet or a door is opened by means other than the key. The turn lights flash and the theft-deterrent horn sounds. When the key is inserted into the door key cylinder and turned to unlock, the warnings stop.

SECURITY AND LOCKS

THEFT-DETERRENT SYSTEM STRUCTURAL VIEW

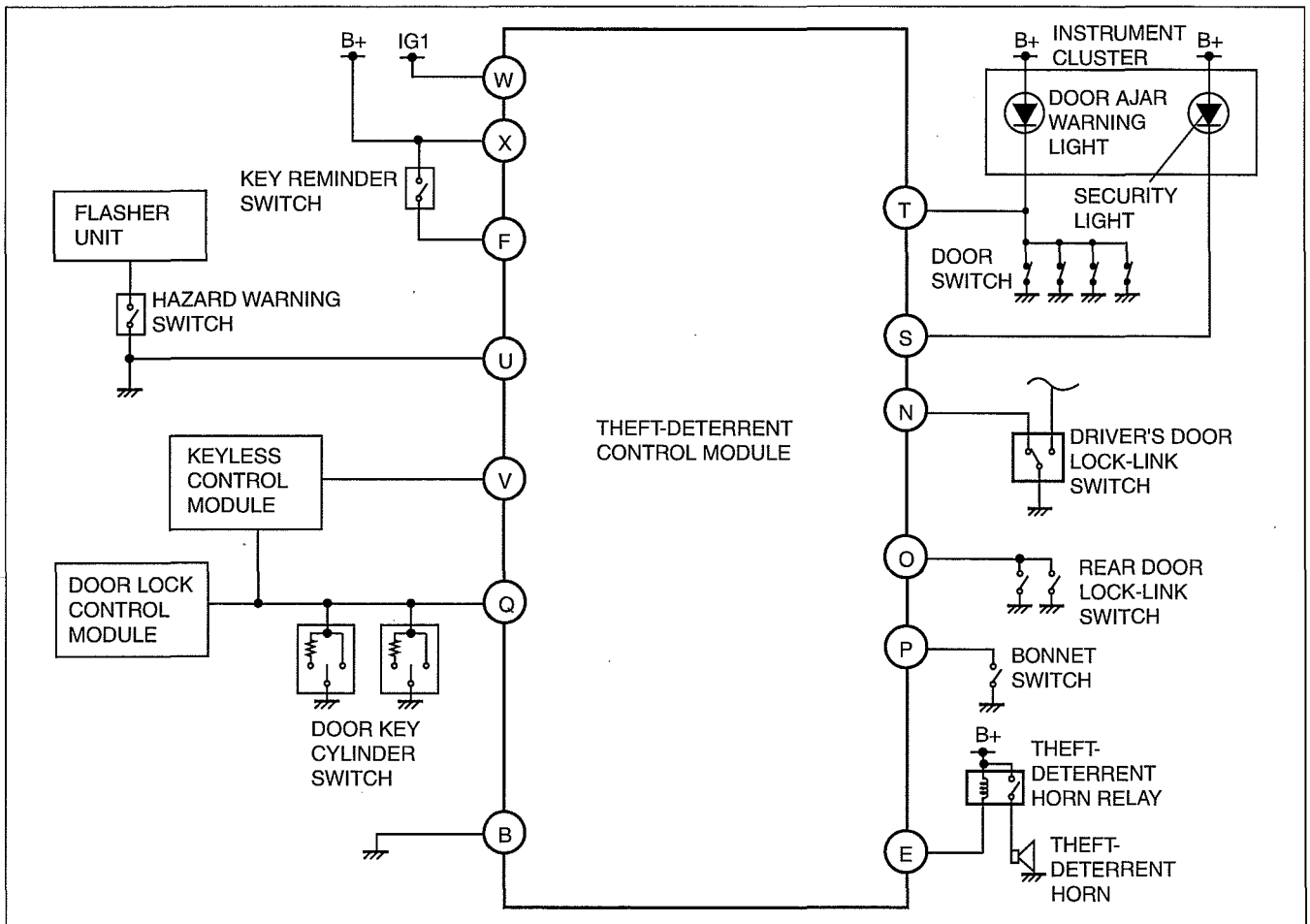
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DBR914ZTB006

THEFT-DETERRENT SYSTEM SYSTEM WIRING DIAGRAM

dcf091450000103

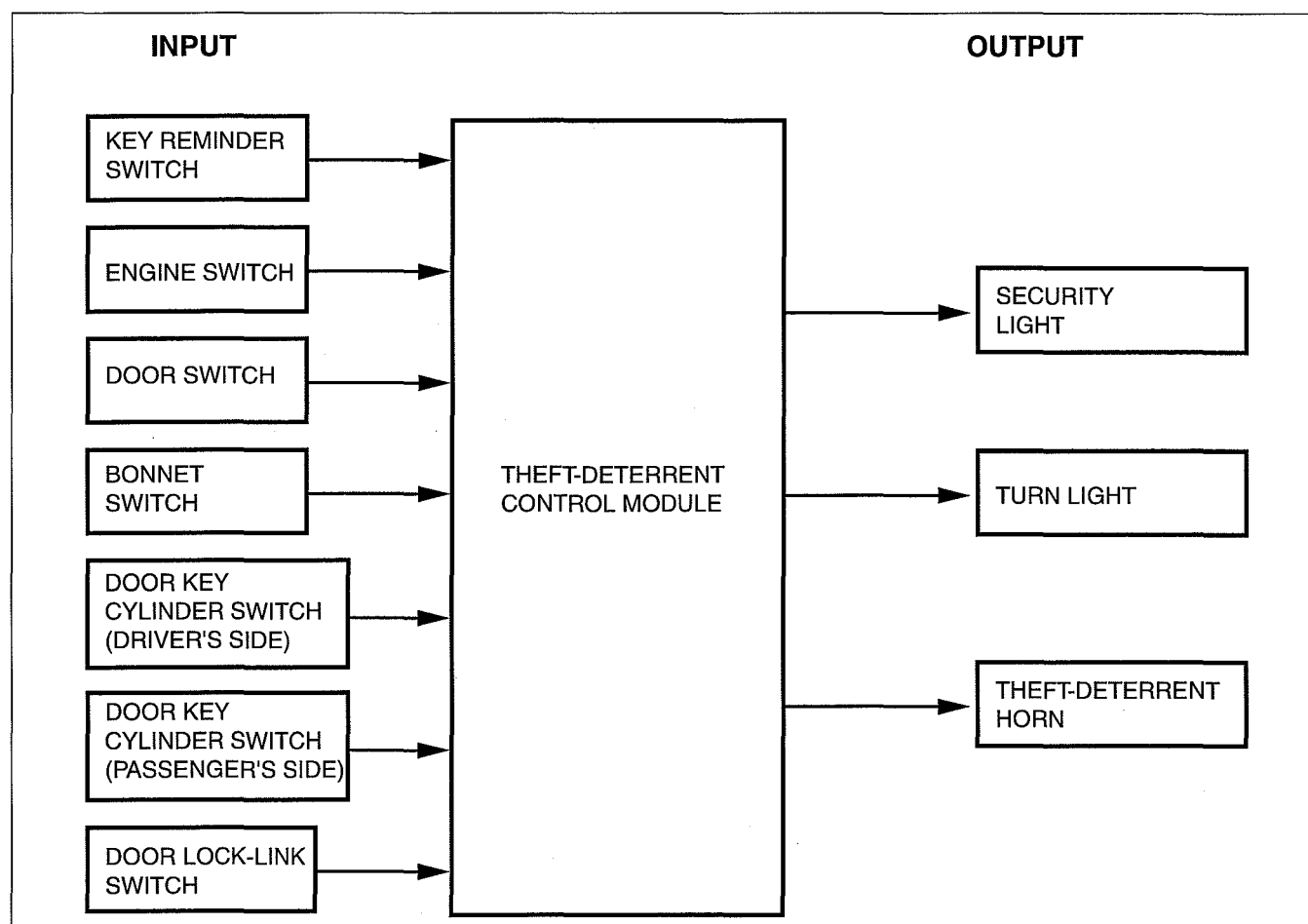


DFC914ZTB111

SECURITY AND LOCKS

THEFT-DETERRENT SYSTEM INPUT/OUTPUT DIAGRAM

dcf091450000104



DBR914ZW003

THEFT-DETERRENT SYSTEM CONDITION

dcf091450000105

System phase		Dead	Initial	Prearming 1	Prearming 2	Arming	Alarm 1	Alarm 2
INPUT	Timer period	—	—	—	—	—	*1	—
	Key reminder switch	On (at least one is on)	Off	Off	Off	Off	—	—
	Engine switch		Off	Off	Off	Off	On or Off (at least one is on)	On or Off
	Door switch	—	Off	On (at least one is on)	On or Unlock (at least one is on or lock)	Off		On or Off
	Bonnet switch	—	Off			Off		On or Off
	Door lock-link switch	—	—	—		Off		On or Off
	Driver's and passenger's door key cylinder switch	—	—	—	—	Off/Lock	Off/Lock	Off/Lock
OUTPUT	Theft-deterrent horn	Off	Off	Off	Off	Off		Off
	Turn light	Off	Off	Off	Off	Off		Off
	Security light	Off	Off	Off	Off		Off	Off

*1: Theft-deterrent horn timer period is 25 s. Turn light timer period is 4.5 min.

SECURITY AND LOCKS

THEFT-DETERRENT SYSTEM PHASE

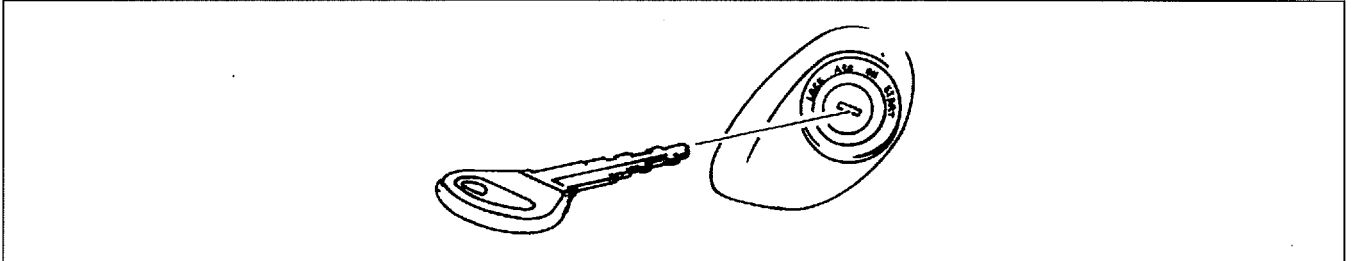
dcf091450000i06

Dead

- The condition before the key is removed from the steering lock. (The key is at either ON position, ACC position, or LOCK position.)
- The security light is not lit at this time.

Initial

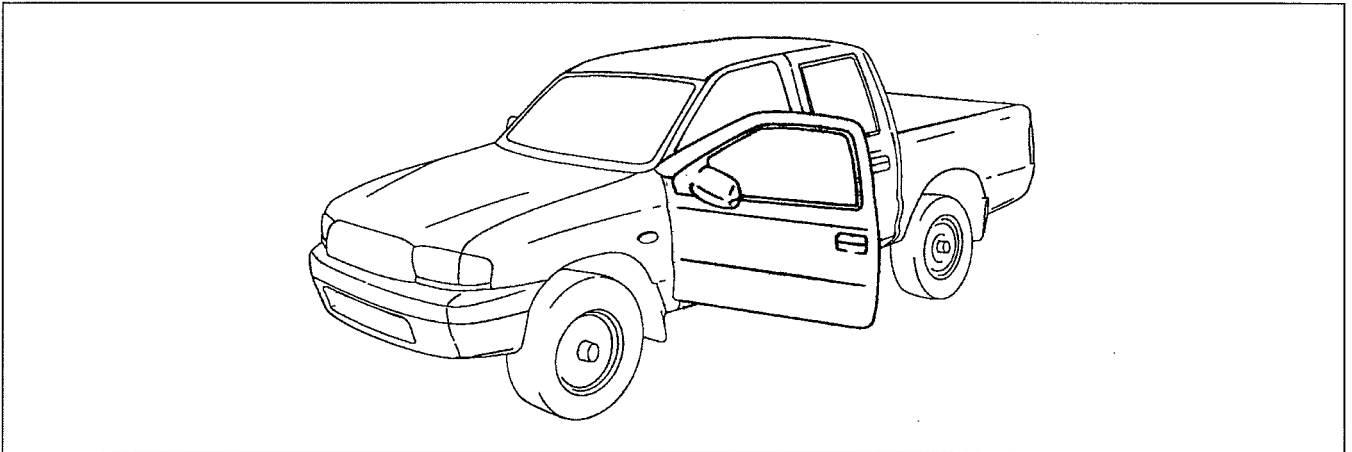
- The condition after the key is removed from the steering lock (with all doors and the bonnet closed).
- The security light is not lit at this time.



ZBR8120W004

Prearming 1

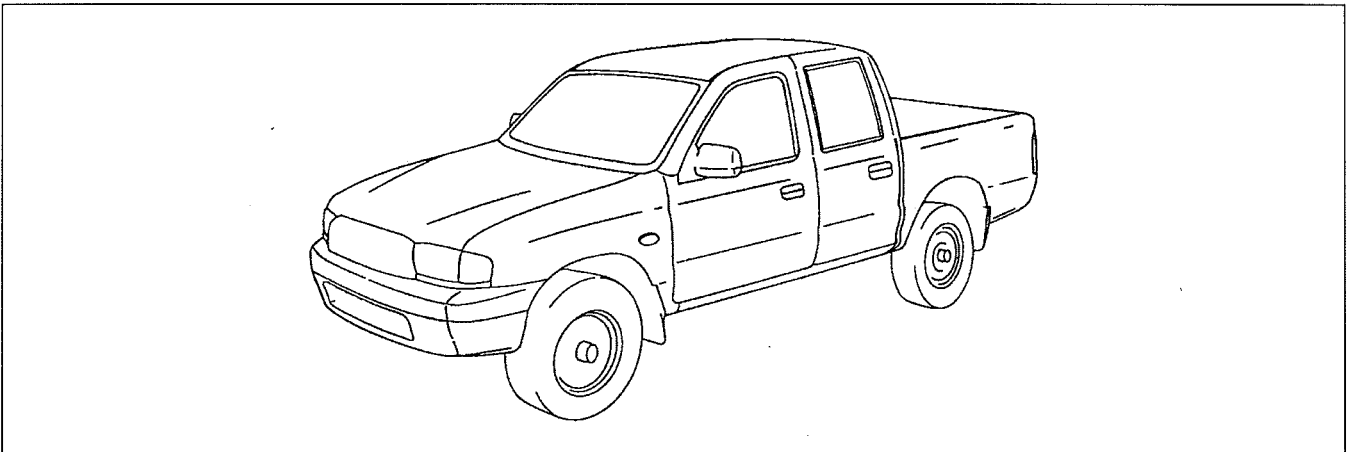
- The key is removed from the steering lock and a door or a bonnet is open.
- The security light is not lit at this time.



ZBR8120W005

Prearming 2

- The key is removed from the steering lock and a door is opened or not locked, and the bonnet is opened.
- The security light is not lit at this time.

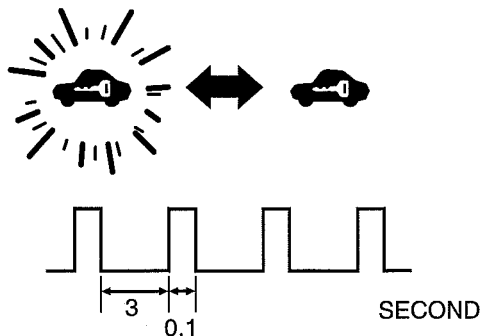


ZBR8120W006

SECURITY AND LOCKS

Arming 1

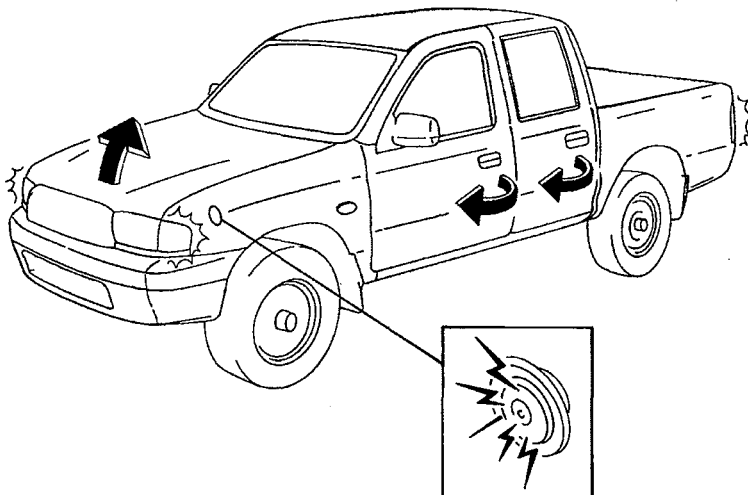
- The doors are locked with the key after all doors and the bonnet are closed.
- The turn lights have been flashed for 2 s shifting to "Arming 1" phase from "Preaming 2" phase.
- The security light is flashes at an interval of 3 s. The alarm system is fully set.



DBR914ZWB004

Alarm 1

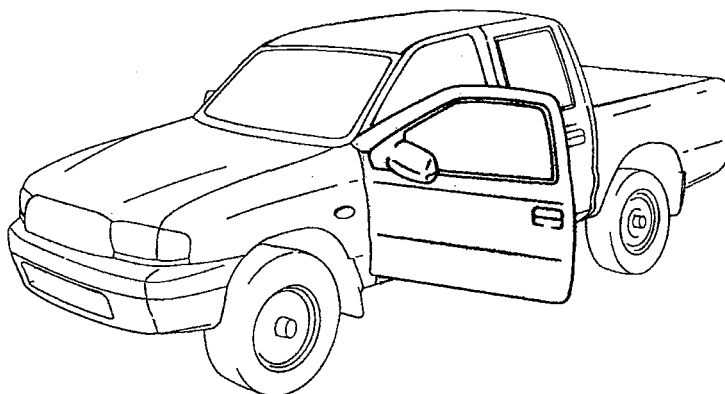
- A door or the bonnet is opened without using the key, or the ignition circuit is short-circuited. The theft-deterrent horn sounds intermittently for 25 s and the turn lights flash intermittently for 4.5 min.
- The theft-deterrent horn sounds intermittently for 25 s and turn lights flash intermittently for 4.5 min again when a door or the bonnet is opened or closed without using the key from the period of 25 s until 4.5 min.



ZBR8120W008

Alarm 2

- The condition (after 4.5 min have passed from the time "Alarm 1" phase was activated) in which the warning system is deactivated.
- When the door or the bonnet is opened again, the condition returns to "Alarm 1" phase.

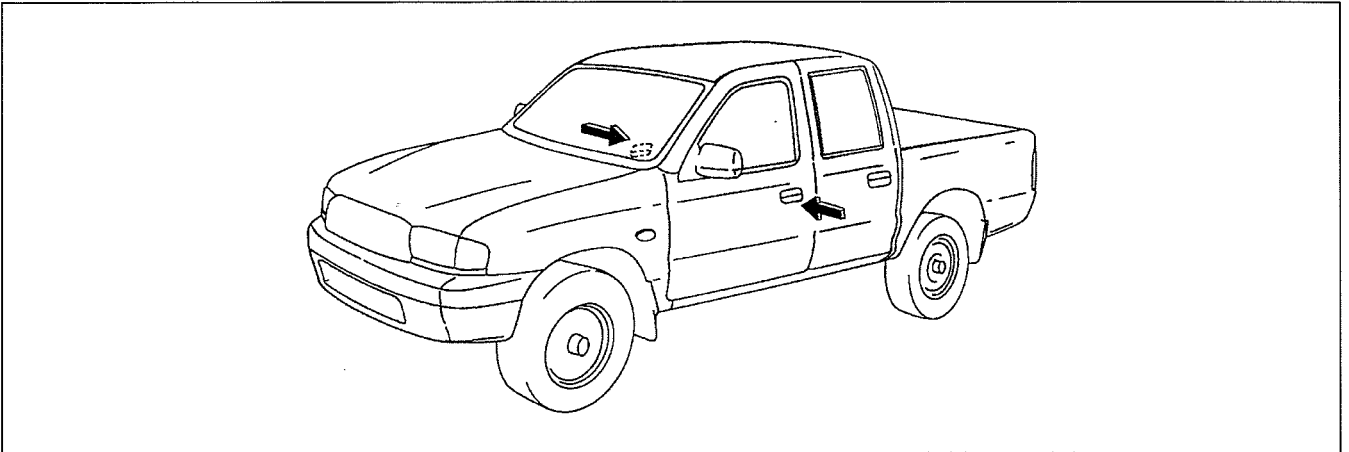


ZBR8120W009

SECURITY AND LOCKS

Alarm Stop Phase (Initial)

- The warning system is cancelled and the condition returns to "Initial" phase when either front door is unlocked with the key.



ZBR8120W010

IMMOBILIZER SYSTEM OUTLINE

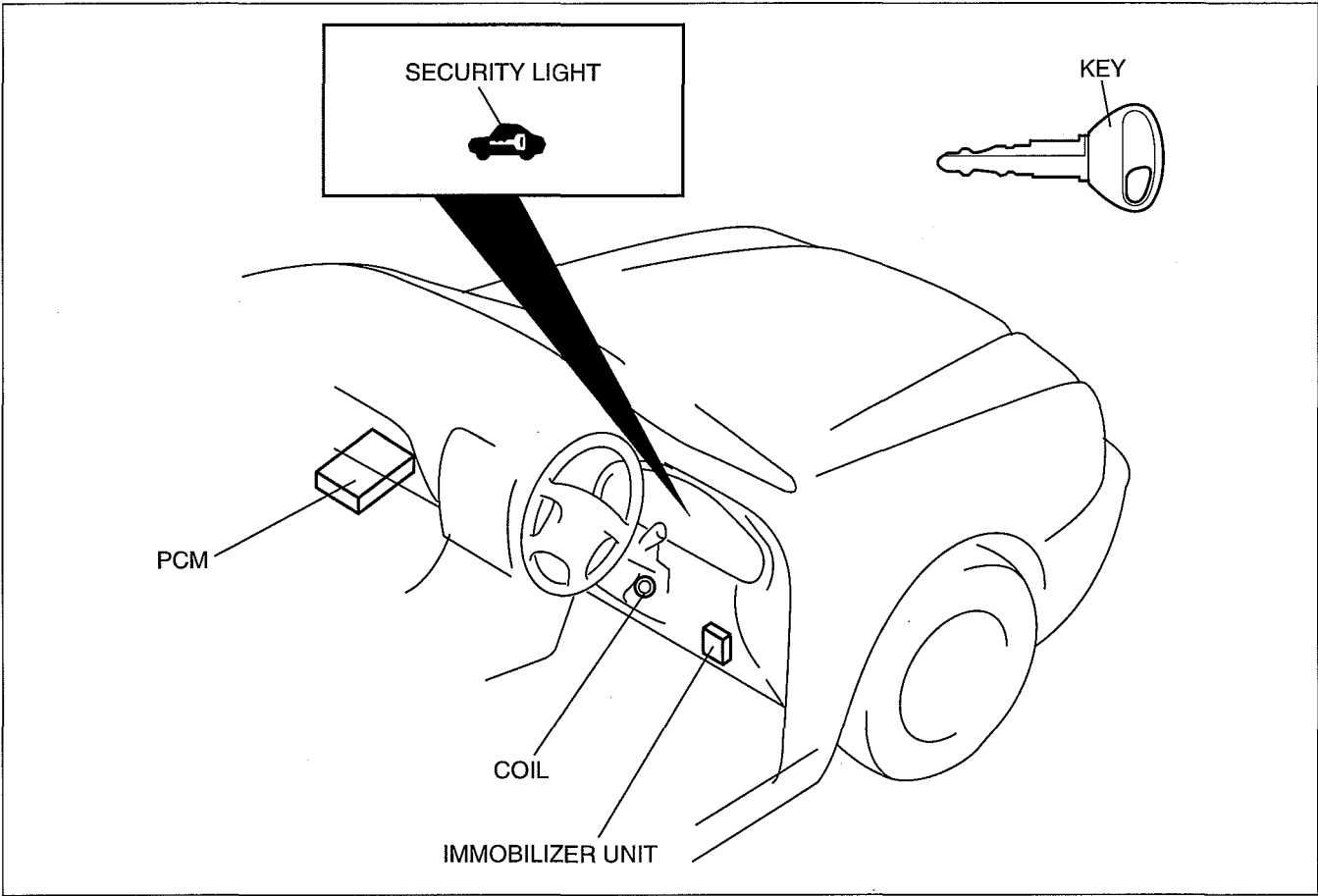
dcf091467000t01

- The immobilizer system prevents the engine from starting even if the engine switch is short-circuited. As a result, it keeps the vehicle from being stolen.
- The system is composed of a special key, the security light, the coil, the immobilizer unit, and PCM. The engine can be started using the key, which has an ID number registered in its transponder and which uses an algorithmic verification method.
- The code word is input to the immobilizer unit and PCM. The vehicle does not start if the code word is not newly input when parts are replaced.
- The procedure for registering the ID number and the procedure for inputting the code word are different according to the parts being replaced and the number of the registered keys.

SECURITY AND LOCKS

IMMOBILIZER SYSTEM STRUCTURAL VIEW

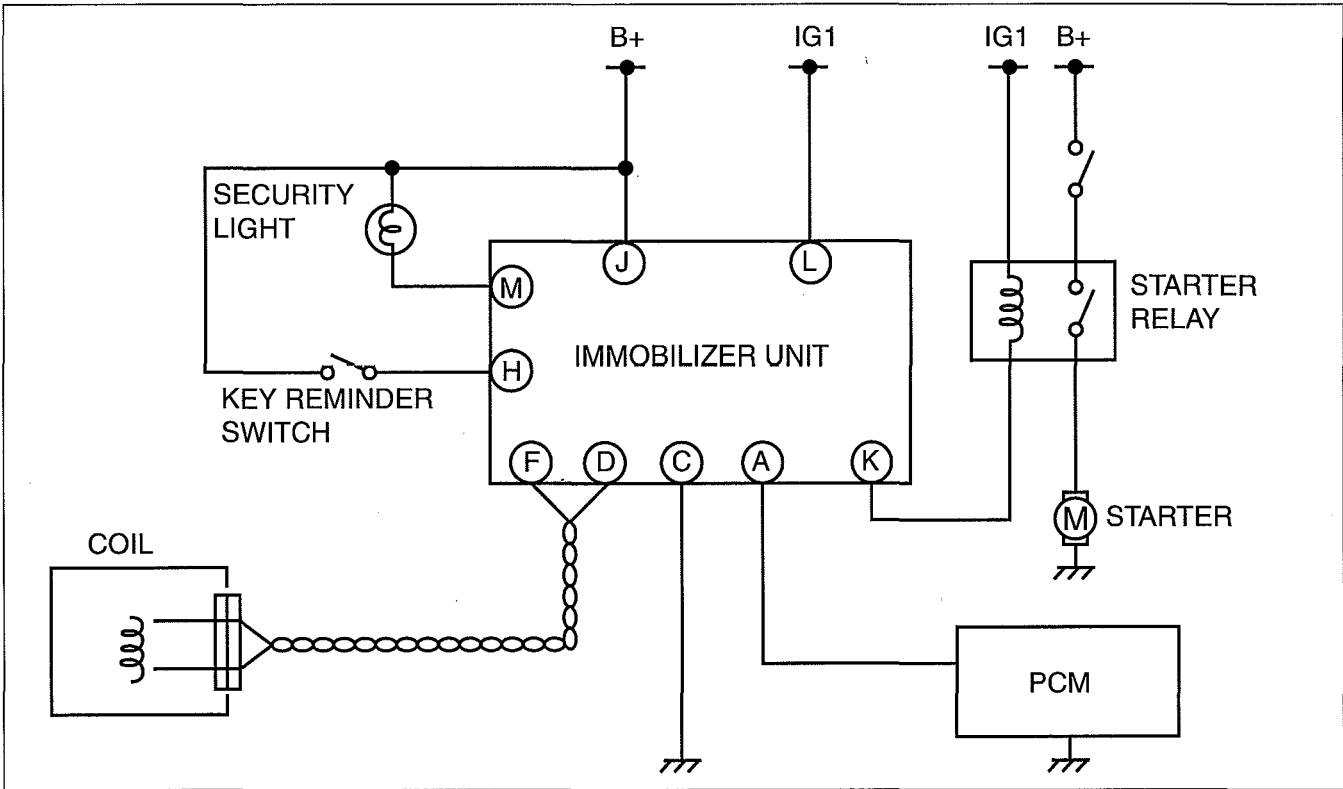
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DBR914ZWB005

IMMOBILIZER SYSTEM WIRING DIAGRAM

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DBR914ZTB007

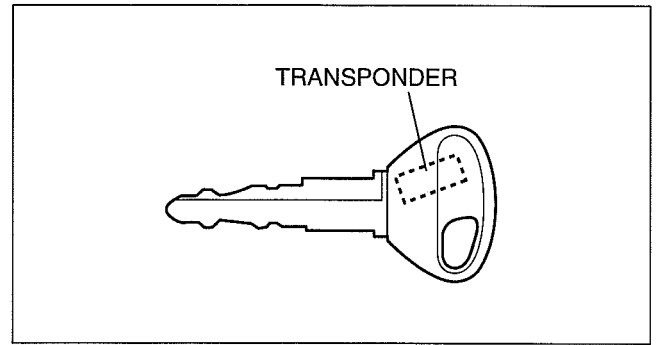
SECURITY AND LOCKS

IMMOBILIZER SYSTEM SYSTEM COMPONENT

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Key

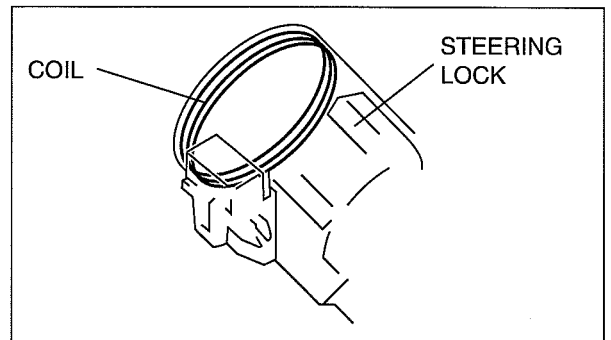
- The transponder in the key has a registered ID number.



DBR914ZTB008

Coil

- The coil is located in the steering lock.
- The coil has an antenna function and sends the ID number to the immobilizer unit.



DBR914ZTB009

Immobilizer Unit

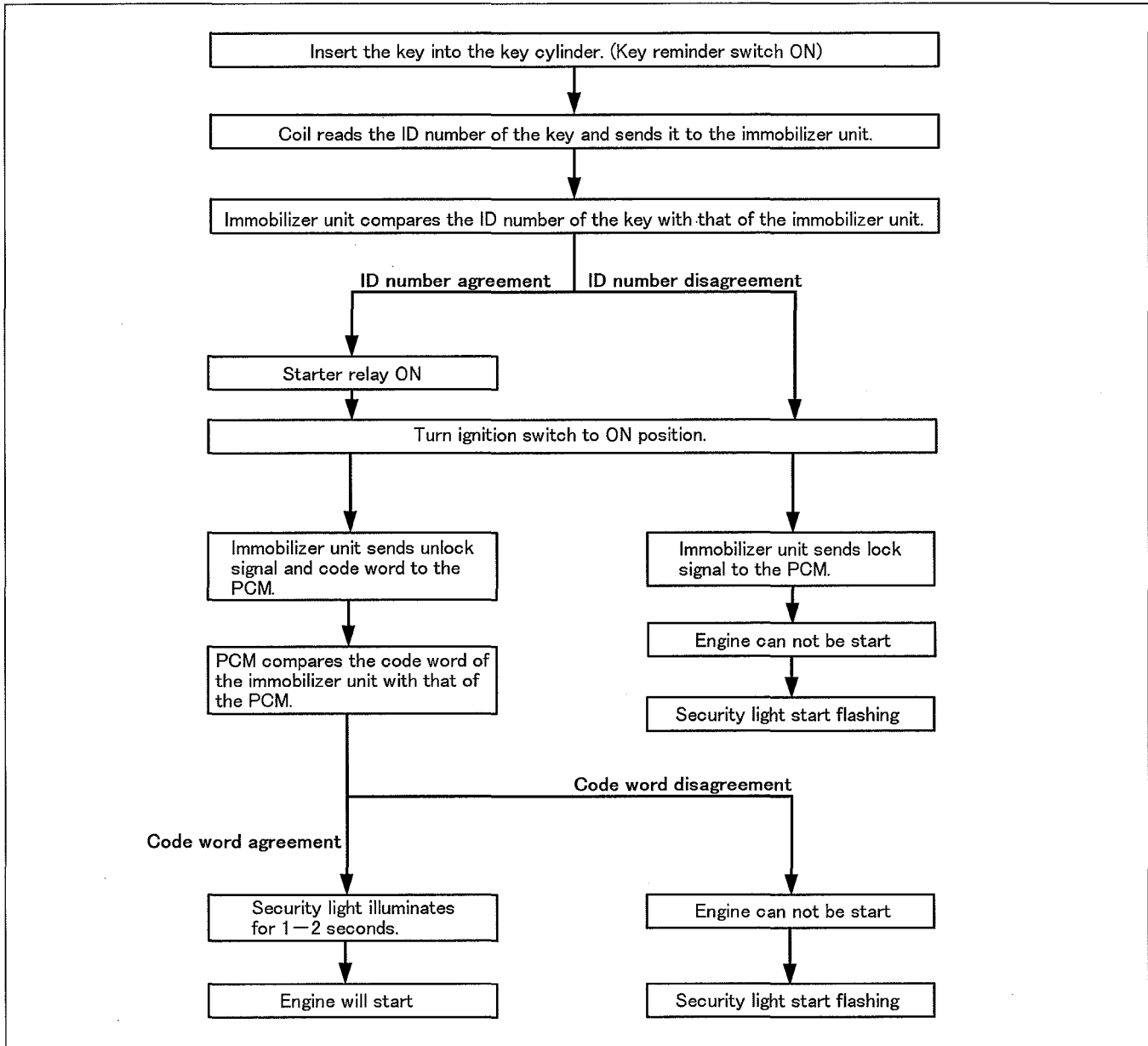
- The immobilizer unit compares the ID number and the algorithmic verification method of the key with those of the immobilizer unit.
- When the ID number and the algorithmic verification method are correct, the immobilizer unit sends the code word to the PCM
- When there is a malfunction in the immobilizer system, the immobilizer unit flashes the security light.

SECURITY AND LOCKS

IMMOBILIZER SYSTEM OPERATION

dcf091467000t05

- This is a conceptual flow chart for understanding how the immobilizer system operates. It shows how the system decides whether or not to start the vehicle when the engine switch is turned to ON position.



DBR914ZTC001

SECURITY AND LOCKS

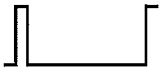
ON-BOARD DIAGNOSTIC SYSTEM [IMMOBILIZER SYSTEM]

dcf091467000i06

Outline

- If the immobilizer system has a malfunction, the on-board diagnostic system displays the diagnostic trouble code number by flashing the security light when the engine switch is at ON position.

DTC Table

DTC	Output pattern	Description
01		ID number unregistered in immobilizer unit is input after engine switch is turned to ON position or cranking engine .
02		ID number format error (voltage range, frequency)
03		ID number is not input into immobilizer unit after engine switch is turned to ON position or cranking engine.
11		Coil or wiring harness between immobilizer unit and coil is open circuit.
21		Code word/ID number stored in immobilizer unit EEPROM cannot be read.
24		Open or short circuit in wiring harness between immobilizer unit and PCM.
30		PCM is defective. (Malfunction of communication line inside PCM)

(1)

(2)

(3)

09-16 EXTERIOR TRIM

EXTERIOR TRIM OUTLINE 09-16-1

EXTERIOR TRIM STRUCTURAL
VIEW 09-16-1

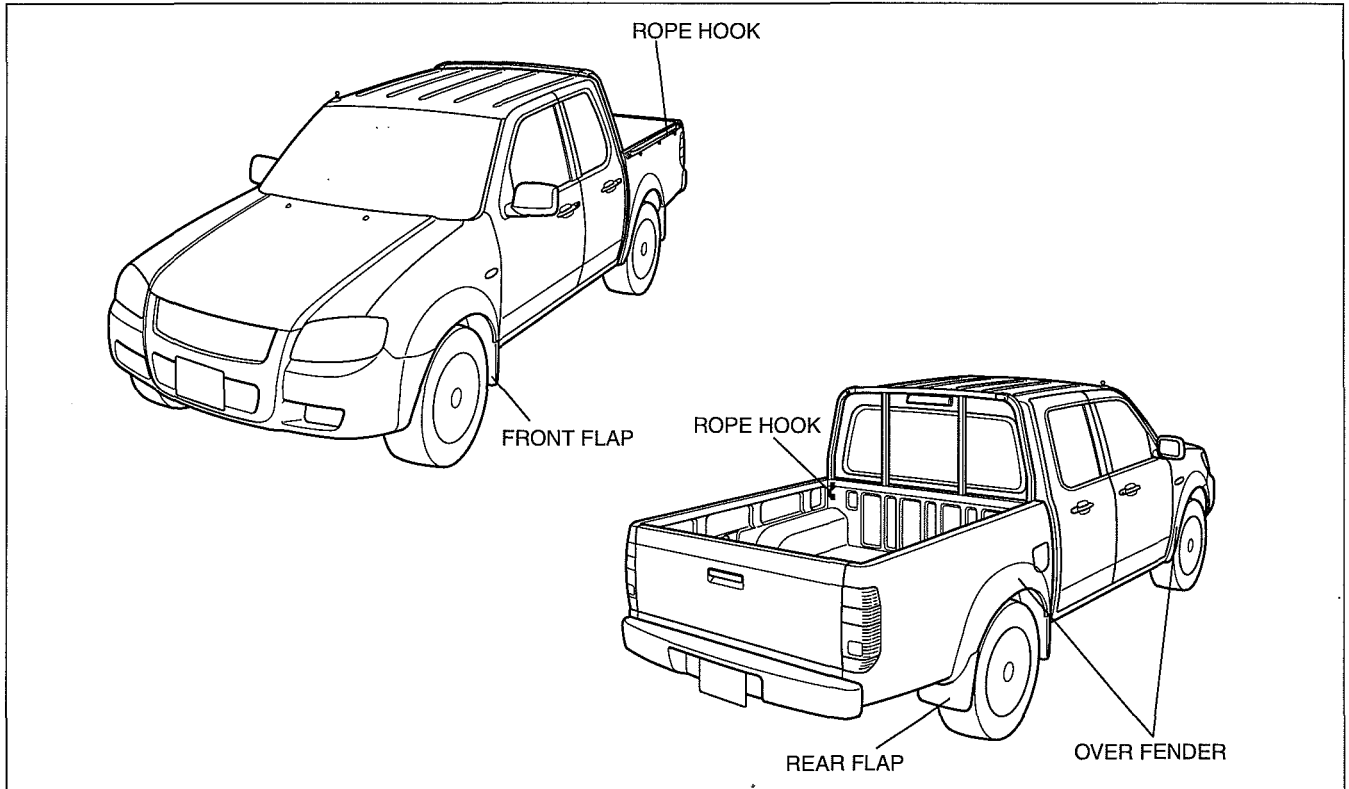
EXTERIOR TRIM OUTLINE

dcf091600000101

- Front and rear flaps have been adopted.
- Over fender has been adopted.
- Rope hook has been adopted.

EXTERIOR TRIM STRUCTURAL VIEW

dcf091600000102



DCG916ZTB000

(1)

(2)

(3)

09-17 INTERIOR TRIM

INTERIOR TRIM OUTLINE 09-17-1

INTERIOR TRIM STRUCTURAL VIEW . . 09-17-1

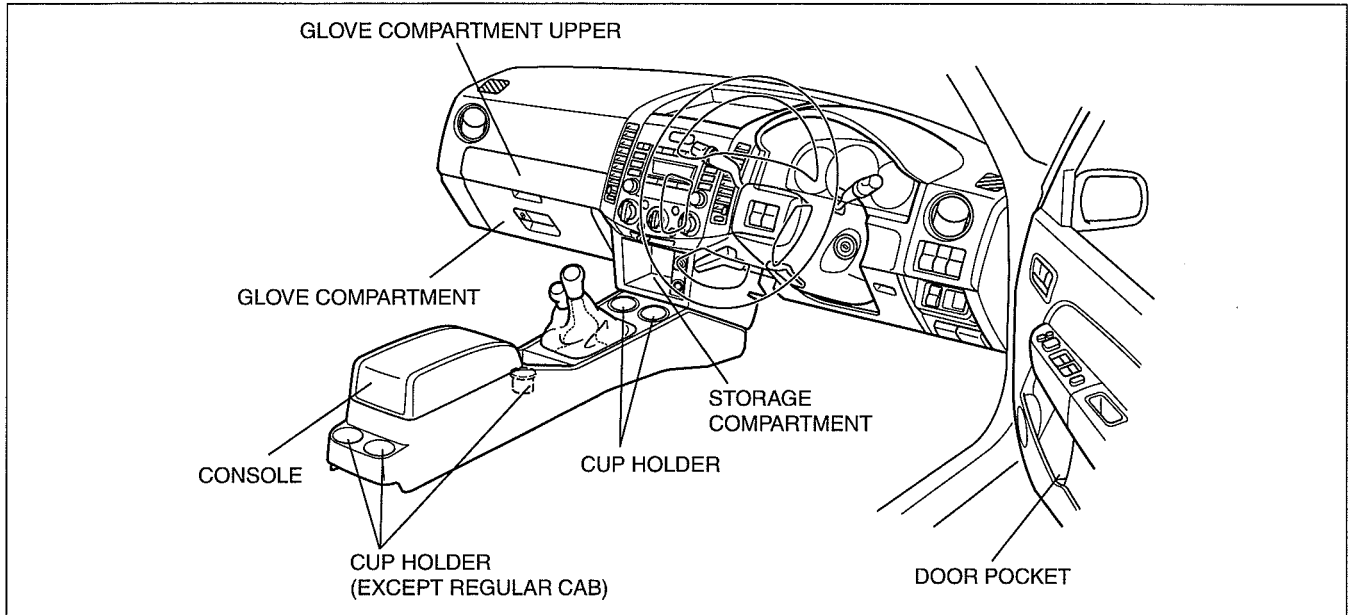
INTERIOR TRIM OUTLINE

dcf091755000101

- Glove compartment upper have been added.
- Various storage spaces have been added.

INTERIOR TRIM STRUCTURAL VIEW

dcf091755000102



DCG917ZTB000

09-18 LIGHTING SYSTEMS

LIGHTING SYSTEM OUTLINE	09-18-1
LIGHTING SYSTEMS SPECIFICATION	09-18-1
LIGHTING SYSTEM STRUCTURAL VIEW	09-18-2

FRONT COMBINATION LIGHT CONSTRUCTION	09-18-3
REAR COMBINATION LIGHT STRUCTURAL VIEW	09-18-4

LIGHTING SYSTEM OUTLINE

dcf091800000t01

- Headlight with built-in front turn and parking light has been adopted.
- A projector type has been adopted to the headlight (low-beam).
- Stepped reflectors have been adopted to the rear combination lights.

LIGHTING SYSTEMS SPECIFICATION

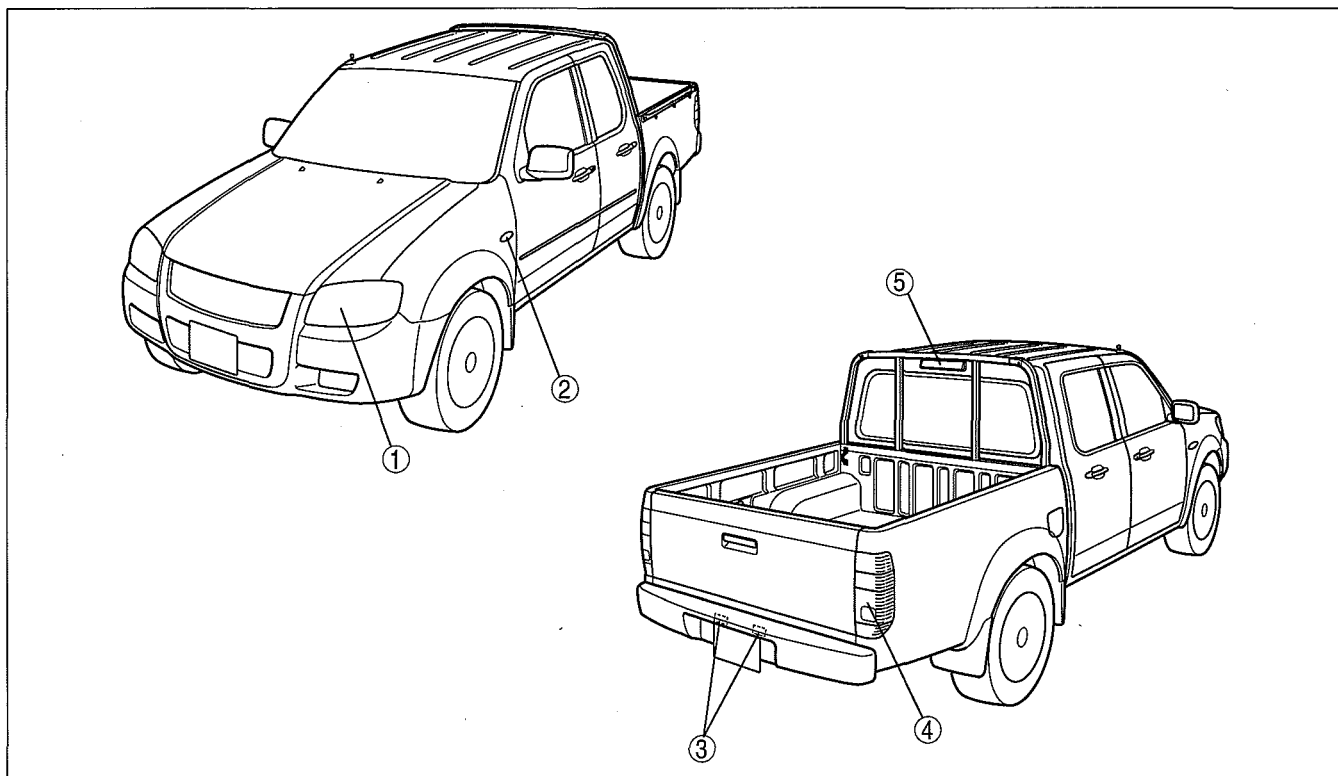
dcf091800000t03

Item			Specifications (W) × number
Exterior light bulb capacity	Headlight bulb	High-beam	60 × 2
		Low-beam	55 × 2
	Front turn light bulb		21 × 2
	Parking light bulb		5 × 2
	Front fog light bulb		55 × 2
	Front side turn light bulb		5 × 2
	Brake/taillight bulb		21/5 × 2
	Rear turn light bulb		21 × 2
	Back-up light bulb		21 × 2
	License plate light bulb		5 × 2
	High-mount brake light bulb		5 × 4
Interior light bulb capacity	Map light bulb		5 × 2
	Interior light bulb		8 × 1
	Glove compartment light bulb		1.7 × 1
	IG key illumination bulb		1.4 × 1
	Console illumination bulb		1.4 × 1

LIGHTING SYSTEMS

LIGHTING SYSTEM STRUCTURAL VIEW

dcf091800000t02

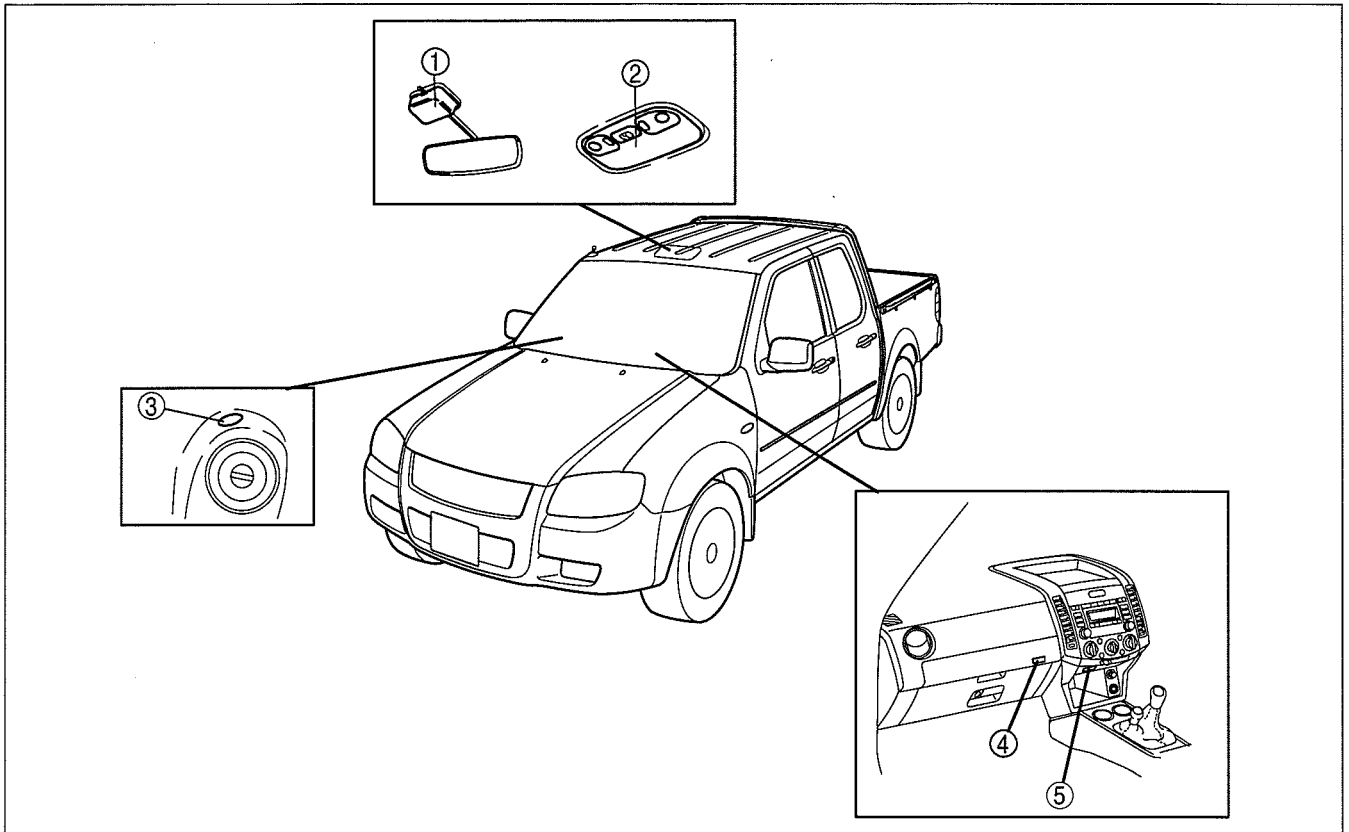


DCG918ZTB000

1	Front combination light
2	Front side turn light
3	License plate light

4	Rear combination light
5	High-mount brake light

LIGHTING SYSTEMS



DCG918ZTB001

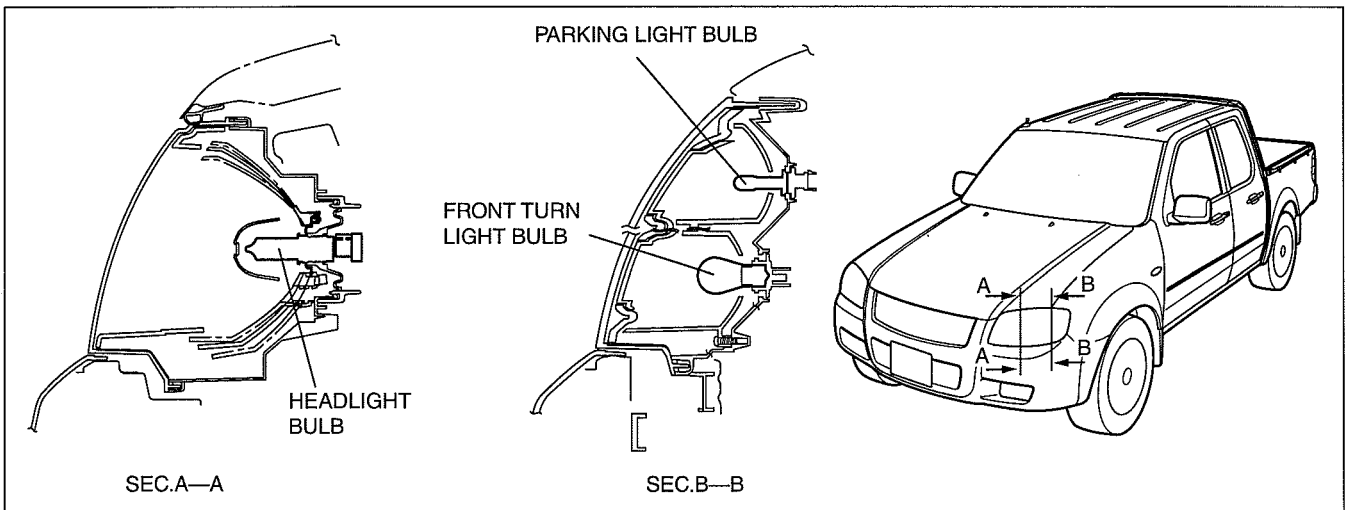
1	Interior light (regular cab)
2	Interior light (except regular cab)
3	IG key illumination

4	Glove compartment light
5	Console illumination

FRONT COMBINATION LIGHT CONSTRUCTION

dcf091851060101

- Projector type headlights (low-beam) have been adopted, and these have been incorporated, along with the front turn light and the parking light, into a single unit to improve size reduction.



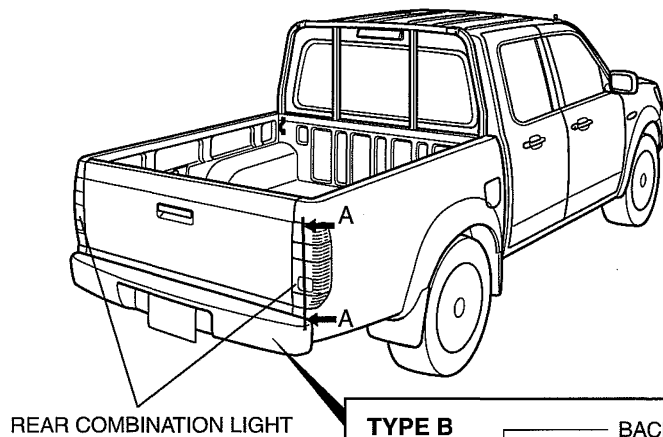
DCG918ZTB002

LIGHTING SYSTEMS

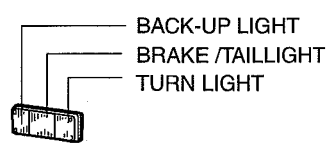
REAR COMBINATION LIGHT STRUCTURAL VIEW

dcf091851150103

TYPE A



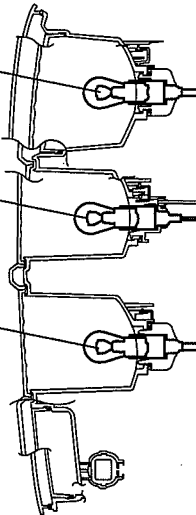
TYPE B



BRAKE
/TAILLIGHT BULB

TURN LIGHT
BULB

BACK-UP
LIGHT BULB



SEC.A—A

DCG918ZTB003

09-20 ENTERTAINMENT

AUDIO SYSTEM OUTLINE.....	09-20-1
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ON-BOARD DIAGNOSTIC SYSTEM FUNCTION	09-20-6
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AUDIO UNIT CONSTRUCTION	09-20-9

AUDIO SYSTEM OUTLINE

dcf092000001101

Type A

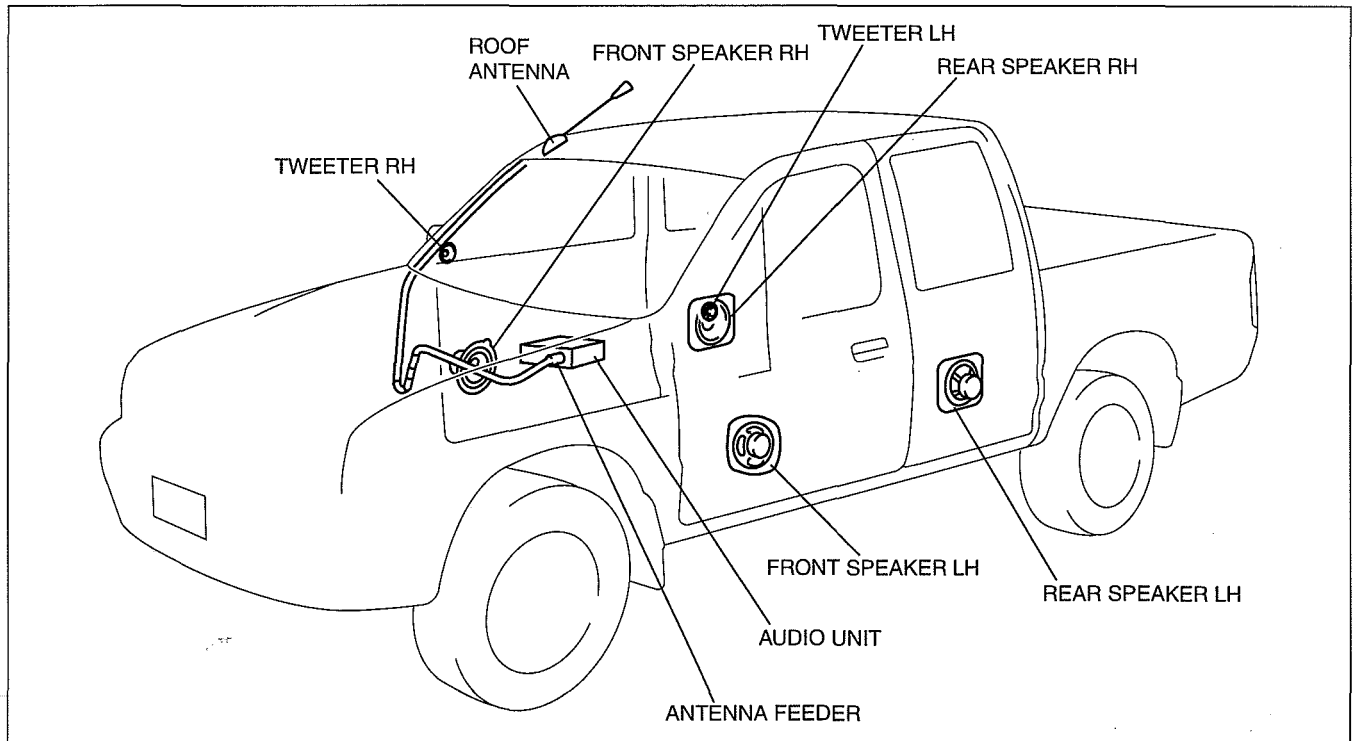
- A center panel unit, composed of the installed audio unit and the audio switches built into the center panel, has been adopted.
- Module availability depends on vehicle grade.
- The auto level control (ALC) function has been adopted.

Type B

- AM/FM radio with the CD player has been adopted.

AUDIO SYSTEM STRUCTURAL VIEW

dcf092000001102



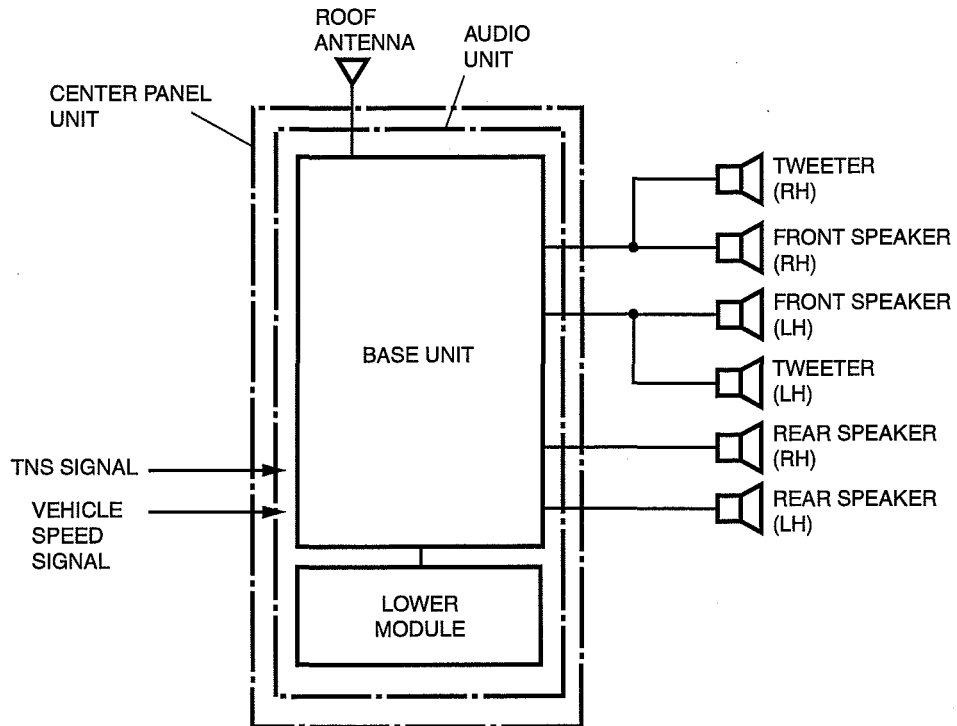
DCF920ZTBA02

ENTERTAINMENT

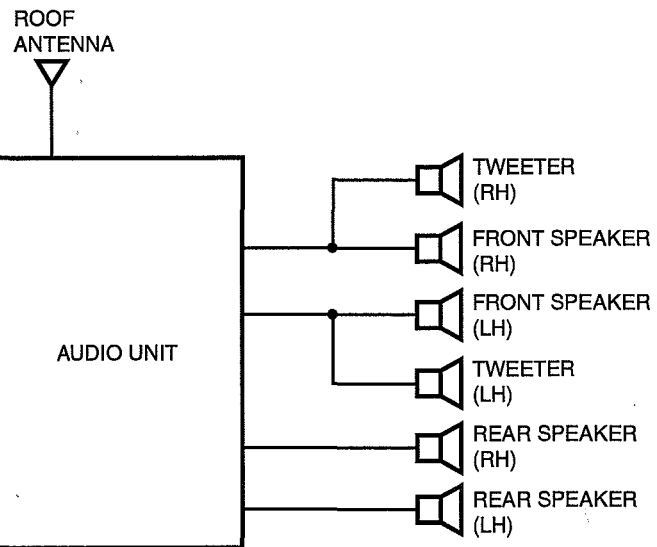
AUDIO SYSTEM BLOCK DIAGRAM

dcf092000001t03

TYPE A



TYPE B



DCF920ZTBA03

ENTERTAINMENT

AUDIO SYSTEM SPECIFICATIONS

id092000100400

Audio Unit

Item			Specification		
			Type A		Type B
			With RDS	Without RDS	
Rated voltage (V)			12		
Frequency band	AM	LW (kHz)	153—279	—	
		MW (kHz)	531—1602	522—1629/ 530—1620	531—1629
	FM (MHz)		87.5—108		
Audio amplifier maximum output power (W)			35×4		
Output impedance (ohm)			4		

Speaker

Item	Specification			
	Front speaker	Rear speaker		Tweeter
		Double cab	Freestyle cab	
Maximum input (W)	25			
Impedance (ohm)	4			
Size (mm)	160	100	160	30

CENTER PANEL UNIT OUTLINE

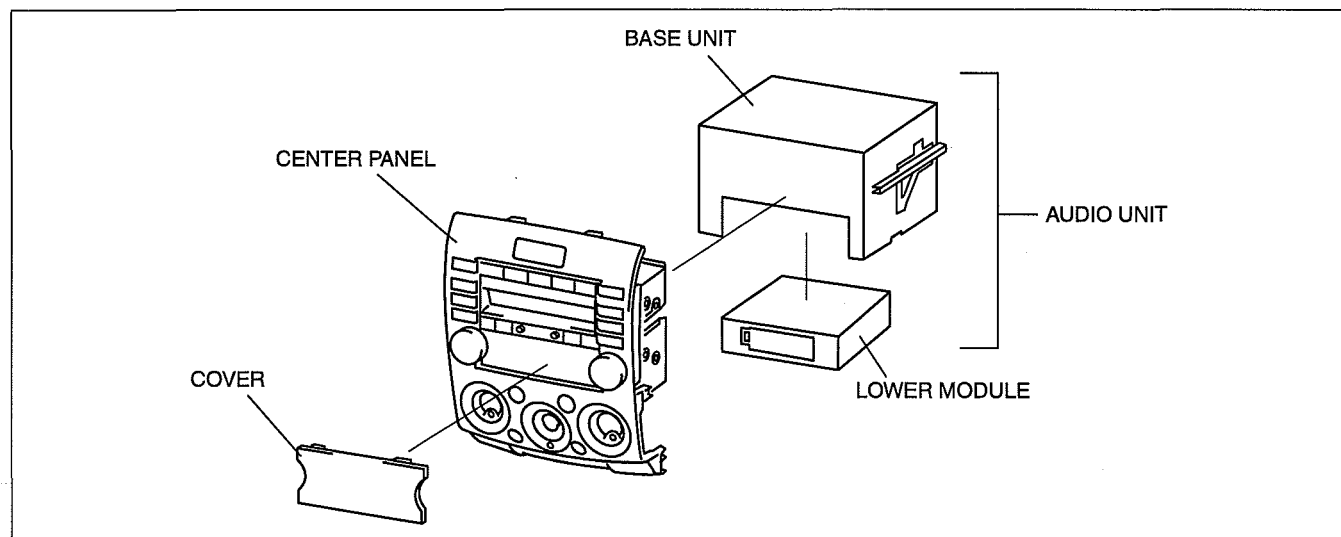
dcf092066900t01

- The center panel unit is composed of the installed audio unit and the audio switches built into the center panel.

CENTER PANEL UNIT CONSTRUCTION

dcf092066900t02

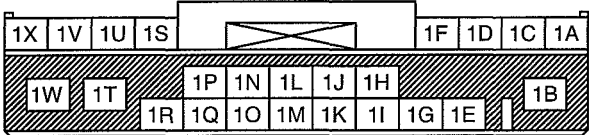
Structural View

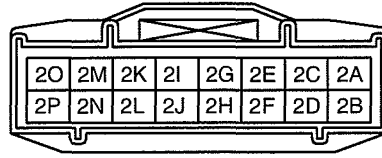


DCF920ZTBA01

ENTERTAINMENT

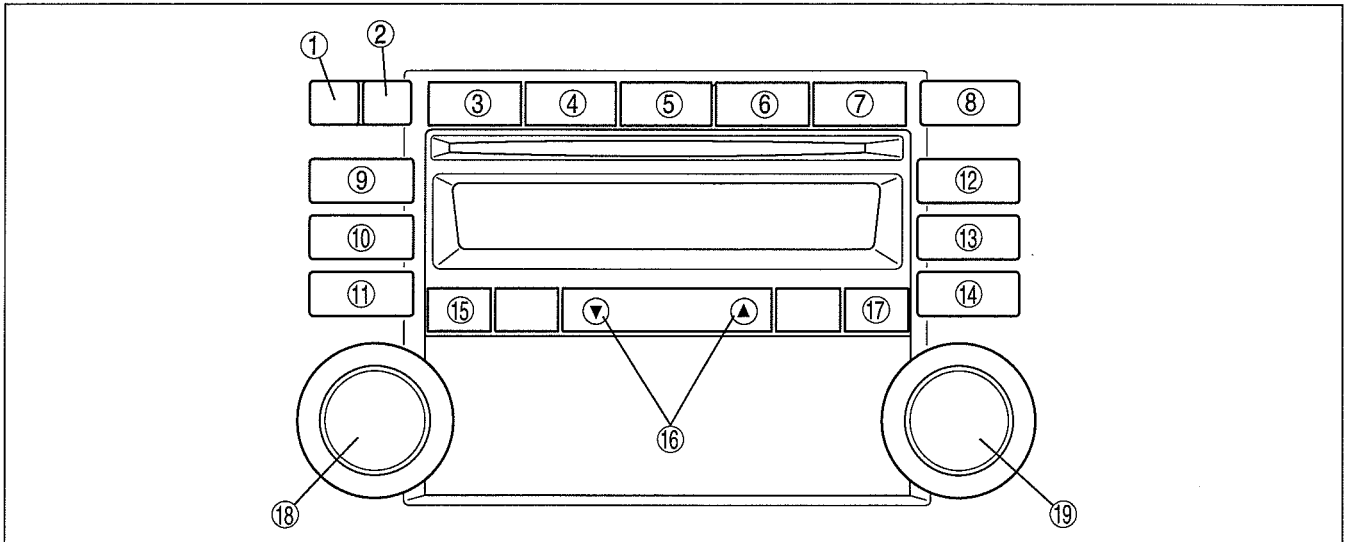
Terminal Layout and Signal Audio unit

Terminal	Signal
	1A Front speaker LH (+)
	1B B+ (Power back up)
	1C Front speaker LH (-)
	1D Front speaker RH (+)
	1E TNS (+)
	1F Front speaker RH (-)
	1G Illumination (-)
	1H Antenna control
	1I Vehicle speed signal
	1J —
	1K UART-1
	1L —
	1M UART-2
	1N —
	1O —
	1P —
	1Q —
	1R ACC
	1S Rear speaker LH (+)
	1T —
	1U Rear speaker LH (-)
	1V Rear speaker RH (+)
	1W Power ground
	1X Rear speaker RH (-)

Terminal	Signal
	2A Power ground
	2B System mute
	2C Input signal RH (+)
	2D Input signal RH (-)
	2E Input signal LH (+)
	2F Input signal LH (-)
	2G Signal ground
	2H TEXT DATA
	2I TEXT CLK
	2J TNS (+)
	2K BUS (-)
	2L BUS (+)
	2M AUX control
	2N —
	2O ACC
	2P +B

ENTERTAINMENT

Button and switch Location



DCF920ZTBS03

1	CLOCK button
2	LOAD button
3	FM 1 button
4	FM 2 button
5	AM button
6	CD button
7	MEDIA button
8	EJECT button
9	Preset button 1
10	Preset button 2

11	Preset button 3
12	Preset button 4
13	Preset button 5
14	Preset button 6
15	DISP/AUTO-M button
16	SEEK/APC/TRACK switch
17	Traffic information button
18	AUDIO CONT/TUNE/TEXT button
19	POWER/VOLUME button

AUTO LEVEL CONTROL (ALC) FUNCTION

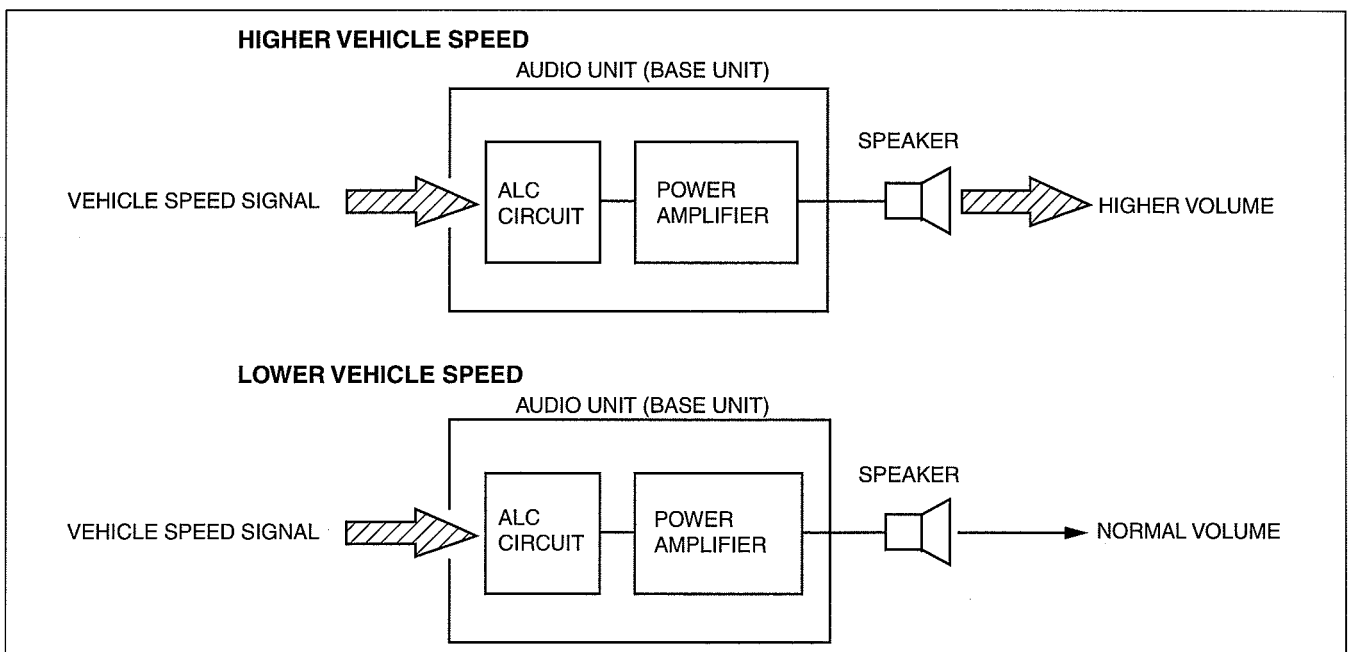
dcf092066900t03

- Adjusts the audio volume so that the sound is balanced against wind and road noise while driving.

AUTO LEVEL CONTROL (ALC) OPERATION

dcf092066900t04

- The audio unit changes the volume automatically based on the vehicle speed signal sent from the instrument cluster.



DCF920ZTBA04

ENTERTAINMENT

- The ALC function is divided into four modes that can be used effectively to match the driving conditions.

Mode	Condition
ALC OFF	ALC function cancelled
ALC LEVEL 1	Outside road noise low
ALC LEVEL 2	Outside road noise slightly high
ALC LEVEL 3	Outside road noise high

ON-BOARD DIAGNOSTIC SYSTEM OUTLINE

dcf092066900t05

- The on-board diagnostic system has a self-diagnostic function and diagnostic assist function to help technicians locate malfunctions.

ON-BOARD DIAGNOSTIC SYSTEM FUNCTION

dcf092066900t06

Self-diagnostic Function

Malfunction detection function

- The malfunction detection section detects malfunctions occurring in the system.

Memory function

- The memory function detects a malfunction, changes it to a DTC, and stores it in the memory. The memory can store a maximum of three DTCs. If another malfunction is detected when three DTCs are already stored, the memory function clears the oldest DTC and stores the new one.
- Once a DTC is stored, it can only be cleared by the designated procedure; not by turning the ignition switch to the LOCK position or disconnecting the negative battery cable. The procedure is mentioned in the Service Section.

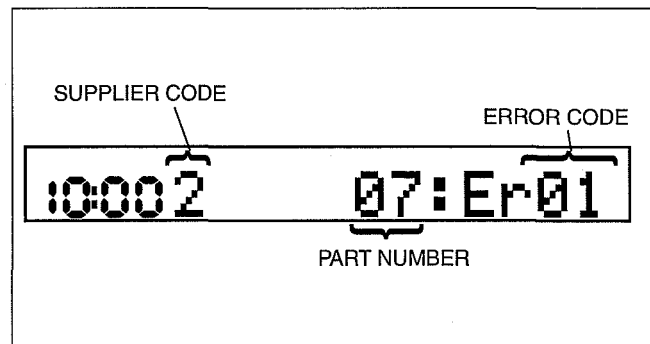
Display function

- When the self-diagnostic function is activated, the information display displays the DTC stored in the memory.
- The DTC consists of the following codes and numbers:
 - Supplier code (indicates manufacturer)
 - Part number (indicates malfunctioning part)
 - Error code (indicates malfunction description)
- Refer to the Service Section for the display method.

Supplier code	Supplier name
1	SANYO Automedia
2	Panasonic
3	Clarion
4	Pioneer

Parts number	Parts name
03	CD player
05	CD changer (external)
06	CD changer (upper module)
07	MD player (lower module)
09	Base unit
10	MP3 applicable CD player
21	Center panel
22	MP3 applicable CD changer

Error code	Malfunction description
01	Internal mechanism error
02	Servo mechanism error
07	Disc reading error
08	Blank media



DCF920ZTBA09

ENTERTAINMENT

Error code	Malfunction description
09	Unplayable files or tracks
10	BUS line (communication line) error
17	Incorrect combination
18	Incorrect combination
19	Communication line
20	Insufficient power supply
21	Amplifier related circuit
22	Tuner error

Screen display		Malfunction location
DTC	Output signal	
03: Er01	—	CD player system
03: Er02	CHECK CD	CD player system
03: Er07	CHECK CD	CD player system
03: Er10	—	CD player communication circuit system
05: Er01	—	CD changer system
05: Er07	CHECK CD	CD changer system
05: Er10	—	CD changer communication circuit system
06: Er01	—	CD changer system
06: Er02	CHECK CD	CD changer system
06: Er07	CHECK CD	CD changer system
06: Er10	—	CD changer communication circuit system
07: Er01	—	MD player system
07: Er02	CHECK MD	MD player system
07: Er07	CHECK MD	MD player system
07: Er08	—	MD system
07: Er10	—	MD player communication circuit system
09: Er20	—	Power supply circuit to base unit
09: Er21	—	Base unit (peripheral circuit for power amplifier)
09: Er22	—	Base unit (peripheral circuit for tuner)
10: Er01	—	MP3 applicable CD player system
10: Er02	CHECK CD	MP3 applicable CD player system
10: Er07	CHECK CD	MP3 applicable CD player system
10: Er09	CHECK CD	MP3 applicable CD data
10: Er10	—	MP3 applicable CD player communication circuit system
21: Er17	—	Center panel system
21: Er18	—	Center panel system
21: Er19	—	Center panel system
22: Er01	—	CD system
22: Er02	—	CD system
22: Er07	—	CD system
22: Er09	CHECK CD	MP3 applicable CD data
22: Er10	—	MP3 applicable CD changer system
no Err	—	No DTCs stored

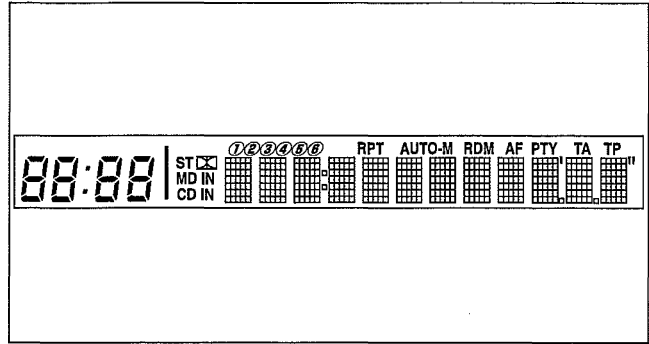
Diagnostic Assist Function

- The diagnostic assist function displays the operating condition of the following functions (components) and forces them to operate in order to examine whether they are malfunctioning or not.
- For the start procedure of each mode, refer to the Service Section.

ENTERTAINMENT

Information display

- The diagnostic assist function illuminates all characters in the information display to check for truncated or faint characters.



DCF902ZWBS04

Speaker

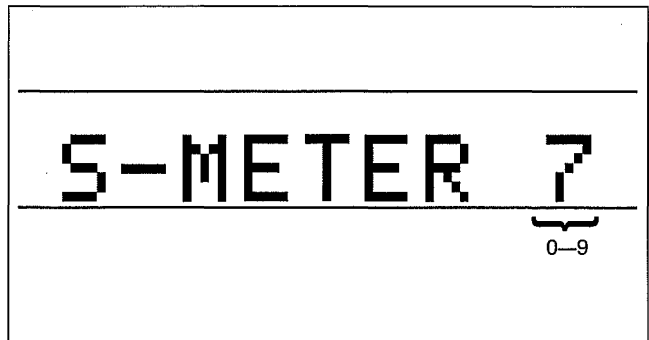
- The diagnostic assist function outputs sound to the speakers in the following order to determine the operating condition of the speakers and wiring harnesses between the base unit and each speaker.

1. Front speaker and tweeter (LH)
2. Front speaker and tweeter (RH)
3. Rear speaker^{*1} (LH)
4. Rear speaker^{*1} (RH)

*1 : May or may not be equipped, depending on the vehicle.

Radio

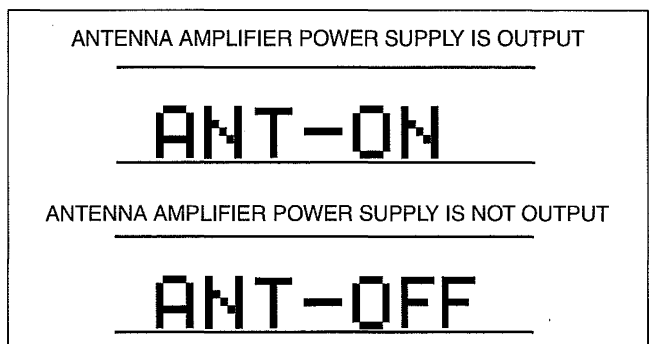
- The diagnostic assist function displays the radio reception condition in 10 levels (0-9) to assist in determining the condition of the antenna, antenna feeders, and base unit (tuner).



E5U920ZS5008

Antenna control condition

- The diagnostic assist function displays the output state of the antenna amplifier power supply to determine the condition of the antenna amplifier, base unit, and wiring harness between the base unit and antenna amplifier.

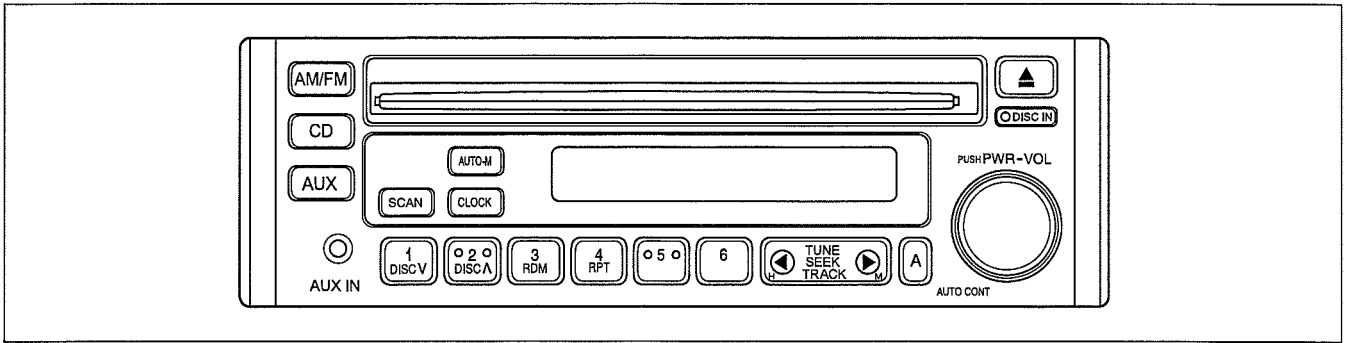


E5U920ZS5202

ENTERTAINMENT

AUDIO UNIT STRUCTURAL VIEW

dcf092066900107



CBF920ZW4001

AUDIO UNIT CONSTRUCTION

Terminal Layout and Signals AM/FM radio with CD player

dcf092066900108

Terminal	Signal
Diagram showing terminal block 1 with terminals labeled 1O, 1M, 1K, 1I, 1G, 1E, 1C, 1A, 1P, 1N, 1L, 1J, 1H, 1F, 1D, 1B.	1A Input RH (+)
	1B Signal ground
	1C Input LH (+)
	1D Combi cont
	1E —
	1F Aux cont
	1G Bus ground
	1H Bus line
	1I ACC
	1J Power ground
	1K B+ (power back up)
	1L System mute (input)
	1M TNS (+)
	1N Illumination (-)
	1O —
	1P —
Diagram showing terminal block 2 with terminals labeled 2M, 2K, 2E, 2C, 2A, 2N, 2L, 2J, 2H, 2F, 2D, 2B.	2A ACC
	2B —
	2C B+ (power back up)
	2D —
	2E TNS (+)
	2F —
	2H —
	2J —
	2K Front speaker LH (+)
	2L Front speaker LH (-)
	2M Front speaker RH (+)
	2N Front speaker RH (-)
Diagram showing terminal block 3 with terminals labeled 3I, 3C, 3A, 3J, 3H, 3F, 3D, 3B.	3A Rear speaker LH (+)
	3B Rear speaker LH (-)
	3C —
	3D —
	3F Rear speaker RH (+)
	3H Rear speaker RH (-)
	3I —
	3J —
Diagram showing terminal block 4 with terminal 4A.	4A Ground (power)

(1)

(2)

(3)

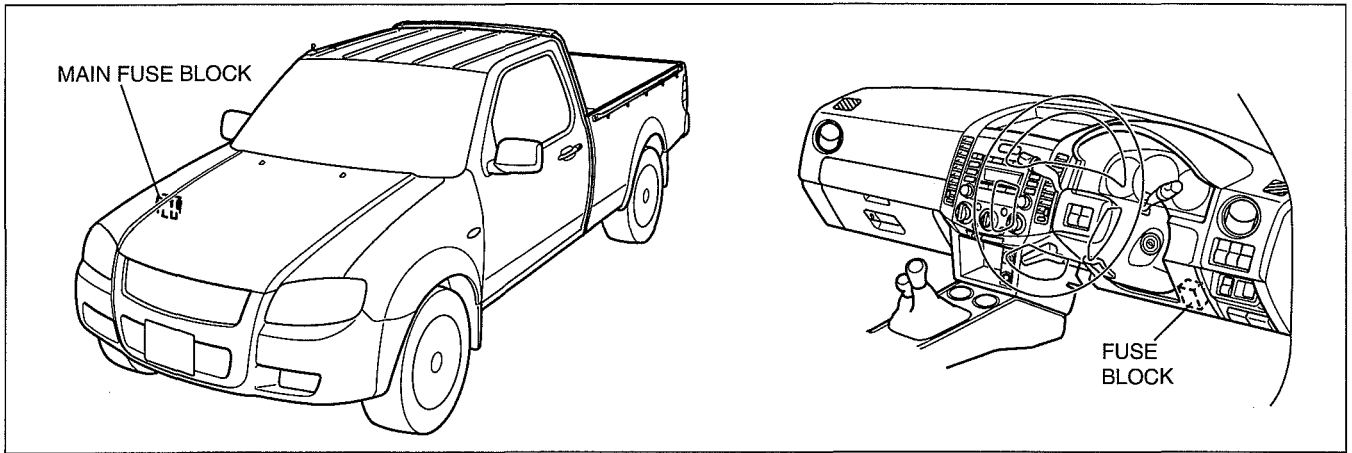
09-21 POWER SYSTEMS

POWER SYSTEMS STRUCTURA

VIEW 09-21-1

POWER SYSTEMS STRUCTURAL VIEW

dcf092167730102



DCF921ZTBO01

09-22 INSTRUMENTATION/DRIVER INFO.

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INSTRUMENT CLUSTER SYSTEM WIRING DIAGRAM	09-22-3
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LIGHTS-ON REMINDER WARNING ALARM CONSTRUCTION/ OPERATION	09-22-6
KEY REMINDER WARNING ALARM OUTLINE	09-22-7

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FUEL GAUGE CONTROL OUTLINE	09-22-8
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WATER TEMPERATURE GAUGE CONTROL OUTLINE	09-22-9
WATER TEMPERATURE GAUGE CONTROL CONSTRUCTION/ OPERATION	09-22-9
HORN CONSTRUCTION	09-22-9

INSTRUMENT CLUSTER OUTLINE

- LEDs have been adopted for warning and indicator lights installed on the instrument cluster.
- A flat-type horn has been adopted.

dcf092255430t01

INSTRUMENT CLUSTER SPECIFICATIONS

Instrument Cluster

dcf092255430t02

Item		Specification
Speedometer	Meter type	Stepping motor type
	Indication range (km/h)	0—180
	Input signal source	Speedometer sensor
	Rated voltage (V)	DC 12
Tachometer	Meter type	Stepping motor type
	Indication range (rpm)	0—5,000
	Red zone (rpm)	4,500—5,000 (WL-3, WL-C) 4,250—5,000(WE-C)
	Input signal source	PCM
	Rated voltage (V)	DC 12
Fuel gauge	Meter type	Stepping motor type (Reset-to-zero type)
	Input signal source	Fuel gauge sender unit
	Rated voltage (V)	DC 12
Water temperature gauge	Meter type	Stepping motor type (Medium range stabilized type)
	Input signal source	Water temperature sender unit (WL-3) ECT sensor (WL-C, WE-C)
	Rated voltage (V)	DC 12
Odometer/ Tripmeter	Display	LCD
	Indication digits	Odometer: 6 digits, Tripmeter: 4 digits
	Rated voltage (V)	DC 12
Warning alarms	Sound frequency (Hz)	2,000—2,200
	Output sound pressure level (dB)	67.5

Clock

Item	Specification
Clock accuracy (Reference value)* (s/day)	-1.5—1.5

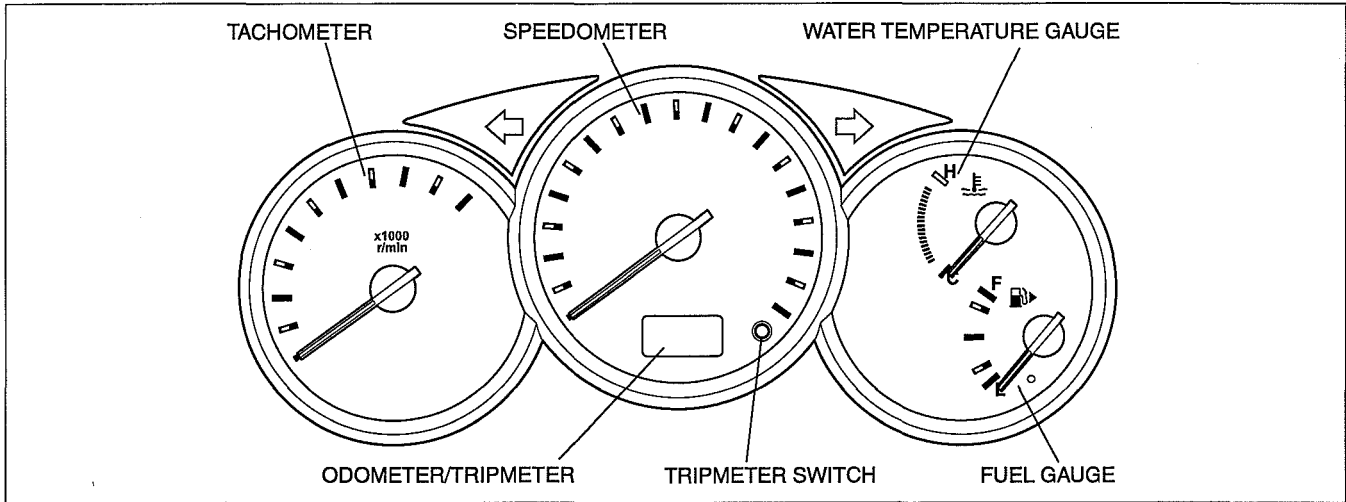
* : If the clock accuracy varies largely from the reference value, battery deterioration or an audio unit (base unit) malfunction may have occurred.

INSTRUMENTATION/DRIVER INFO.

INSTRUMENT CLUSTER STRUCTURAL VIEW

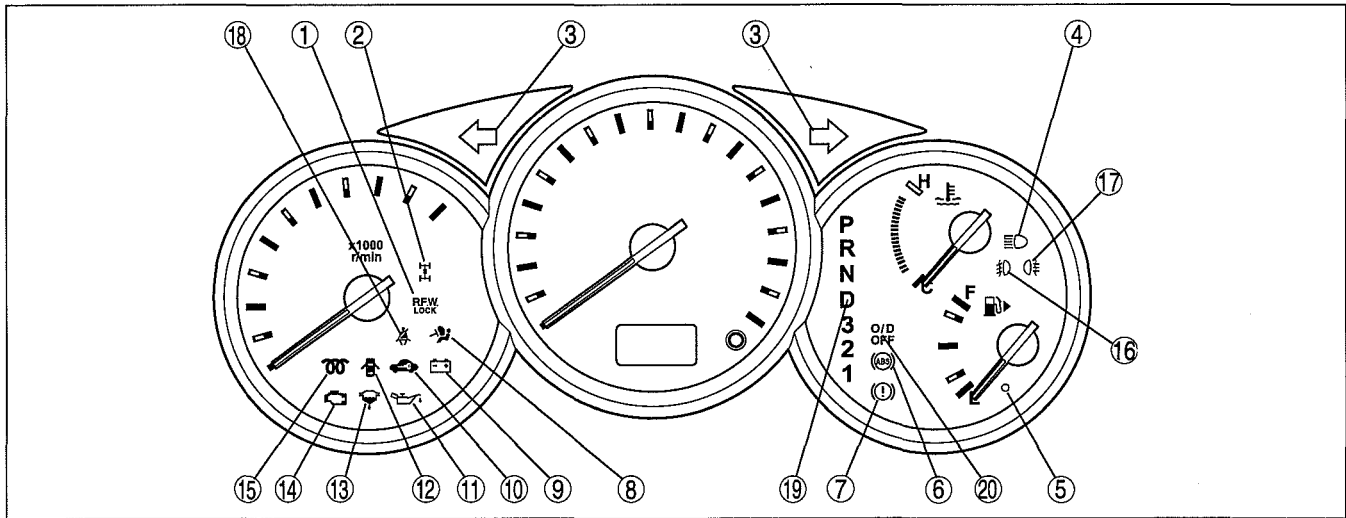
id092200101200

Meter And Gauge



arnffn00000036

Warning And Indicator Light



arnffn000000420

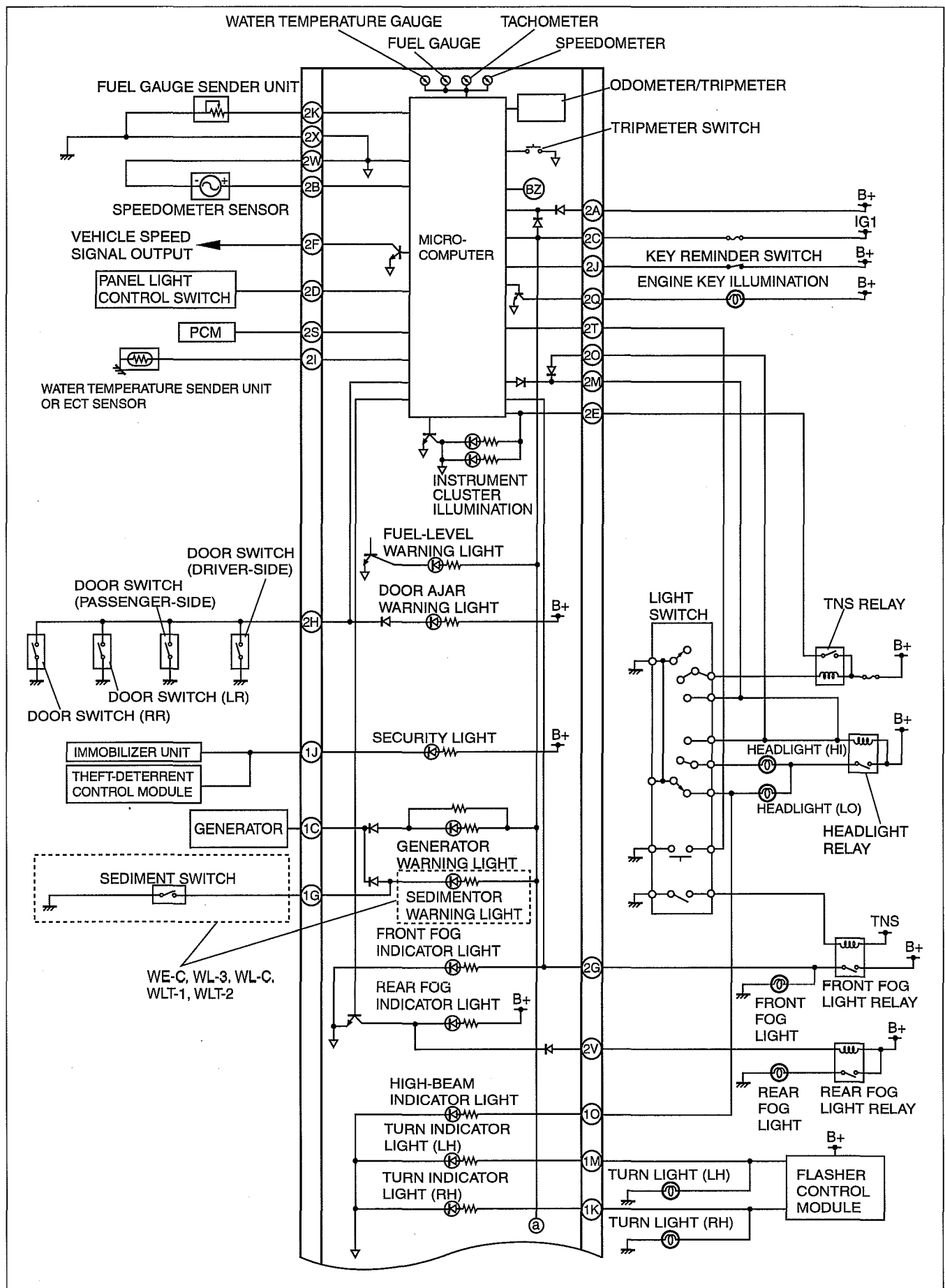
INSTRUMENTATION/DRIVER INFO.

No.	Warning and indicator light	Input signal source	Note
1	RFW indicator light	RFW control module	4x4
2	4x4 indicator light	RFW main switch	4x4
3	Turn indicator light	Flasher control module	—
4	High-beam indicator light	Headlight (HI)	—
5	Fuel-level warning light	Fuel gauge sender unit	—
6	ABS warning light	• ABS HU/CM (with 4-W ABS) • ABS control module (with rear ABS)	With ABS
7	Brake system warning light	• Parking brake switch • Brake fluid level sensor • ABS HU/CM (with 4-W ABS) • ABS control module (with rear ABS)	—
8	Air bag system warning light	SAS control module	With air bag system
9	Generator warning light	Generator	—
10	Security light	• Immobilizer unit • Theft deterrent control module	—
11	Oil pressure warning light	Oil pressure switch	—
12	Door ajar warning light	Door switches	—
13	Sedimentor warning light	Sediment switch	WE-C, WL-3, WL-C, WLT-1, WLT-2
14	MIL	PCM	—
15	Glow indicator light	PCM	WE-C, WL-3, WL-C, WLT-1, WLT-2
16	Front fog indicator light	Front fog light relay	With front fog light
17	Rear fog indicator light	Rear fog light relay	—
18	Seat belt warning light	Buckle switch	—
19	Selector indicator light	TCM	AT
20	O/D OFF indicator light	TCM	AT

INSTRUMENTATION/DRIVER INFO.

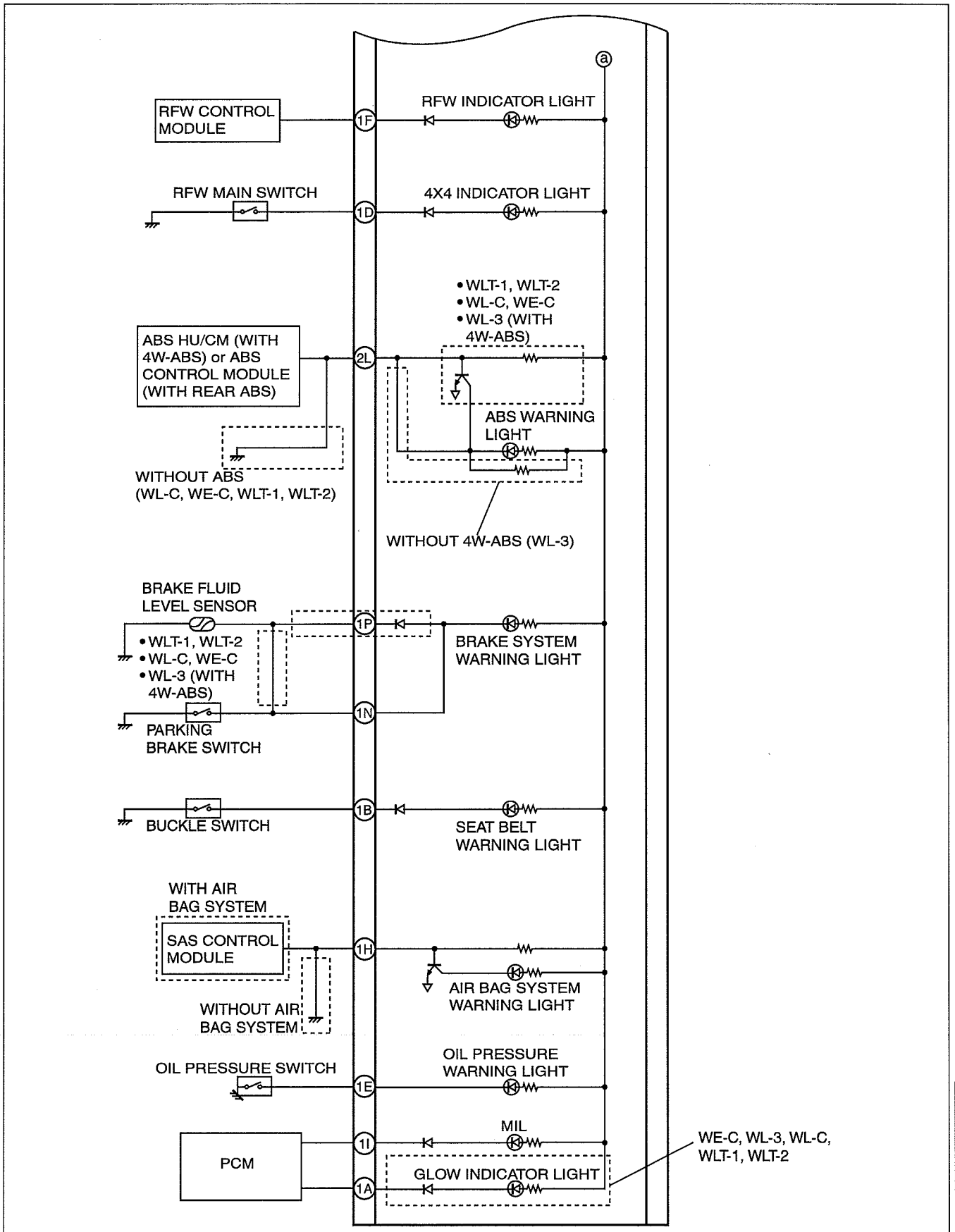
INSTRUMENT CLUSTER SYSTEM WIRING DIAGRAM

id092200101300



arnffn00000062

INSTRUMENTATION/DRIVER INFO.

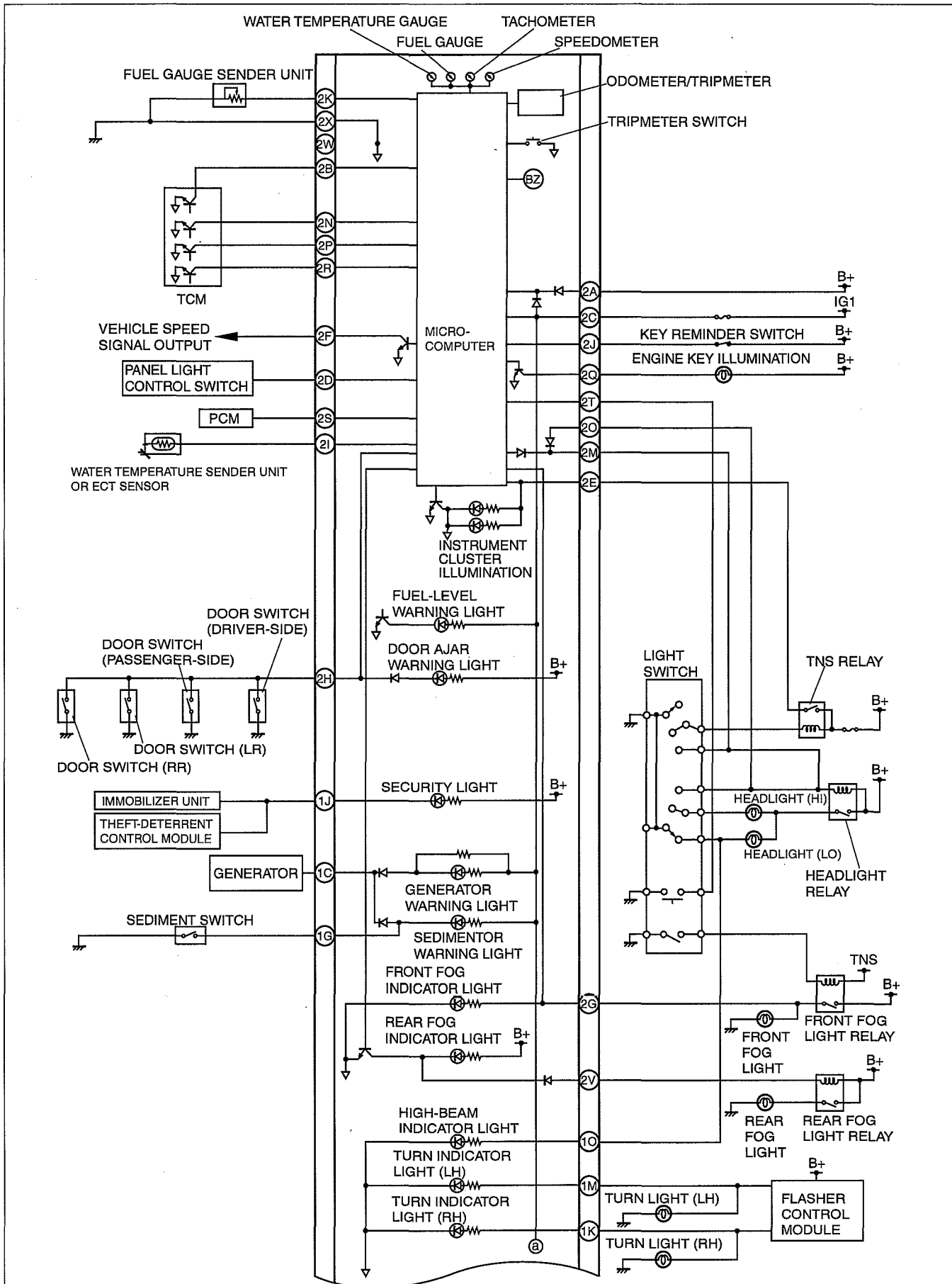


arnfn00000063

INSTRUMENTATION/DRIVER INFO.

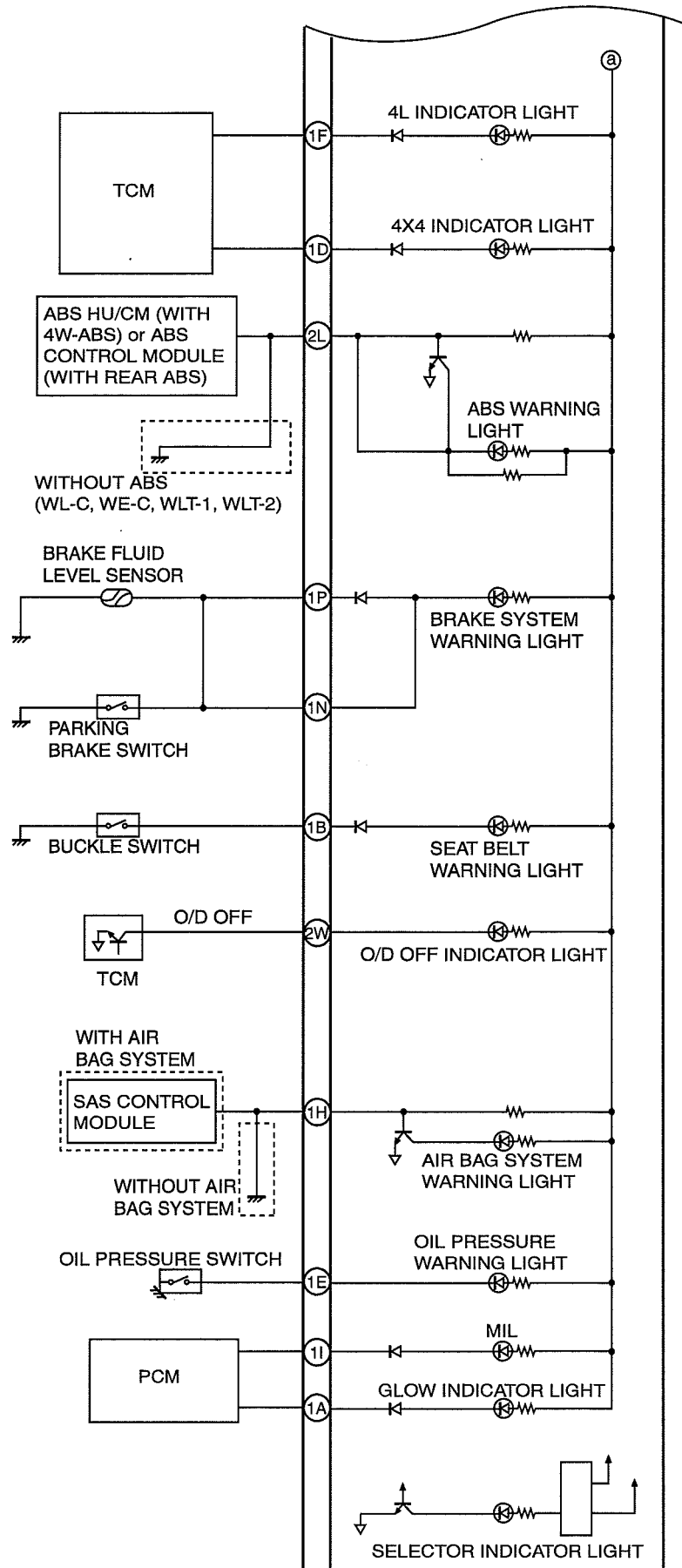
INSTRUMENT CLUSTER SYSTEM WIRING DIAGRAM[5R55S]

Id0922001013z3



arnffn00000293

INSTRUMENTATION/DRIVER INFO.



arnffn00000294

INSTRUMENTATION/DRIVER INFO.

INPUT/OUTPUT CHECK MODE OUTLINE

dcf092255430t05

- The microcomputer built into instrument cluster detects the quality of input signals and individual parts.
- Input/output check mode has both input circuit inspection and individual part inspection functions.

INPUT/OUTPUT CHECK MODE OPERATION

dcf092255430t06

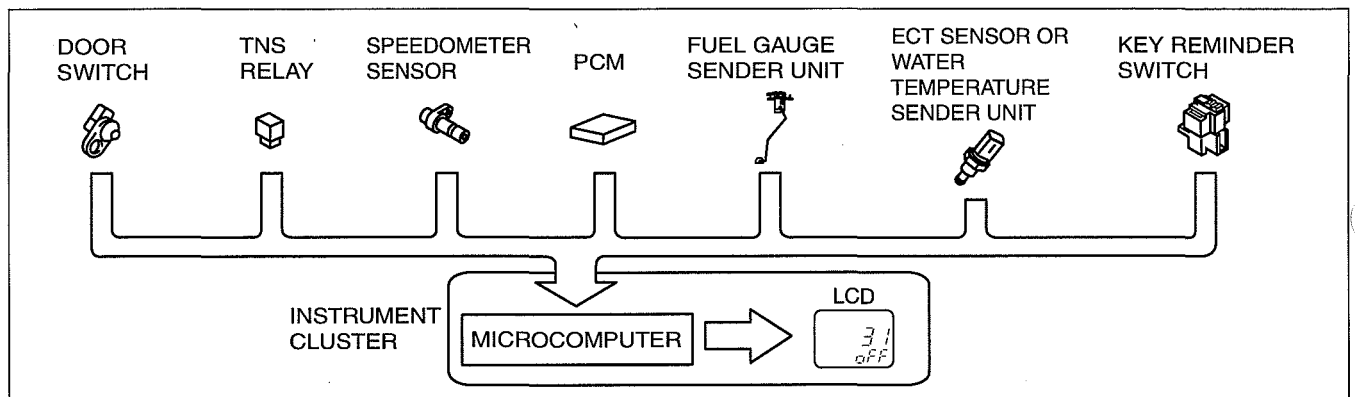
Operation procedure

- Refer to Workshop Manual.

Input circuit check

- When the parts listed in the chart are operated and a signal is output to the instrument cluster, the built-in microcomputer determines the operability of the input circuit based on that signal.

Check code	Parts sending input signal
04	Door switch
08	TNS relay
10	Vehicle speed signal
11	Engine speed signal
22	Fuel gauge sender unit
24	• ECT sensor • Water temperature sender unit
31	Key reminder switch



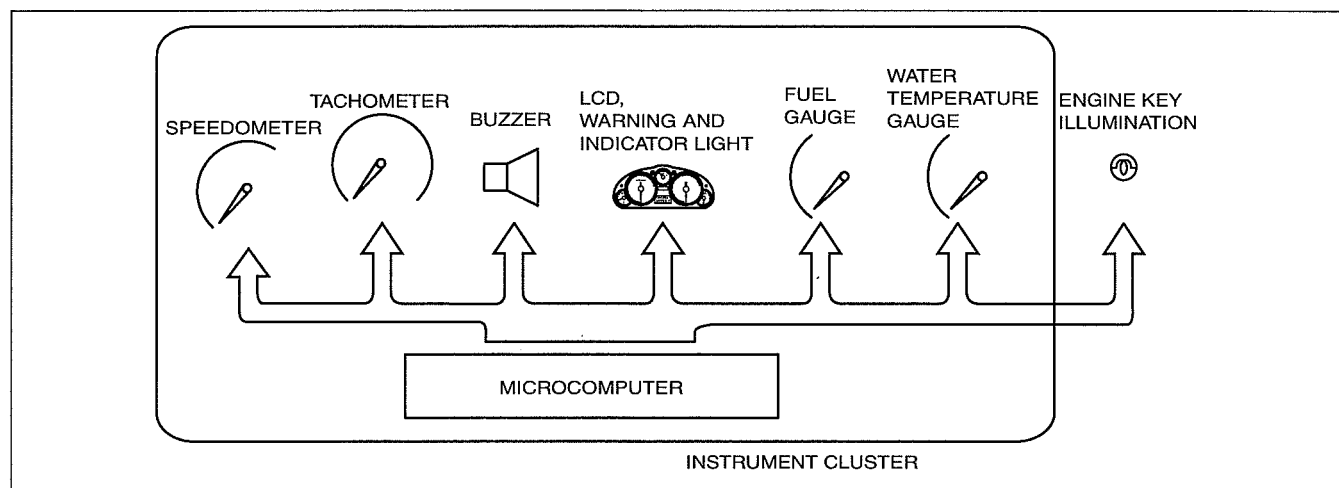
DCF922ZTB005

INSTRUMENTATION/DRIVER INFO.

Individual circuit check

- By operating the parts listed in the chart, the built-in microcomputer determines the operability of the individual parts.

Check code	Parts sending input signal
12	Speedometer
13	Tachometer
14	Buzzer
16	Fuel-level warning light
18	Engine key illumination output
23	Fuel gauge
25	Water temperature gauge
26	LCD, warning and indicator light



DCF9222TBO06

LIGHTS-ON REMINDER WARNING ALARM OUTLINE

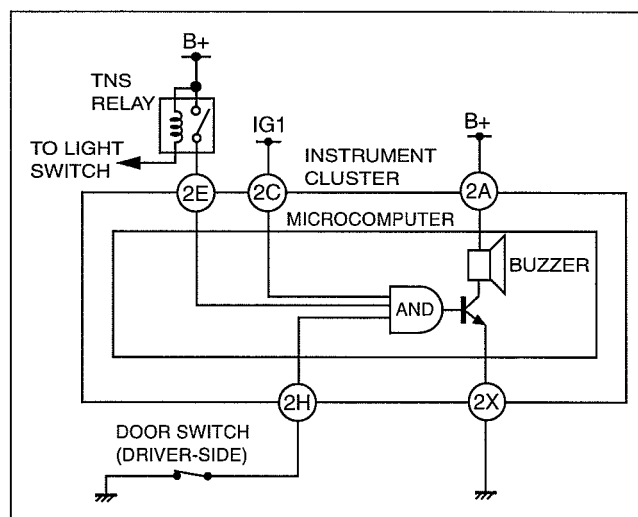
- Warns the driver that the headlights or TNS are on when the driver-side door is opened.

dcf092255430t07

LIGHTS-ON REMINDER WARNING ALARM CONSTRUCTION/OPERATION

dcf092255430t08

System Wiring Diagram



DCF9222TBO07

Operation

- The buzzer in the instrument cluster sounds continuously when all the following three conditions are met:
 - The engine switch is in the LOCK or ACC position.
 - The headlight switch is in the TNS or headlight position.
 - The driver-side door is open (driver-side door switch is on).

INSTRUMENTATION/DRIVER INFO.

KEY REMINDER WARNING ALARM OUTLINE

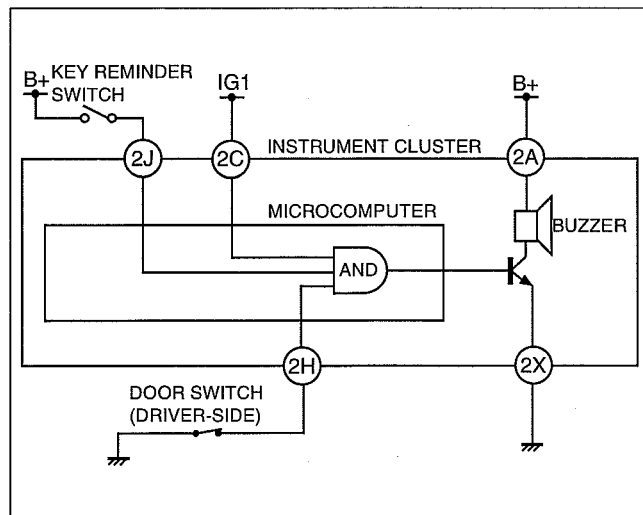
dcf092255430t11

- Warns the driver that the key is in the steering lock when the driver-side door is opened.

KEY REMINDER WARNING ALARM CONSTRUCTION/OPERATION

dcf092255430t12

System Wiring Diagram



DCF922ZTB008

Operation

- The buzzer in the instrument cluster sounds when all the following three conditions are met:
 - The engine switch is in the LOCK or ACC position.
 - The key is in the steering lock (key reminder switch is on).
 - The driver-side door is open (driver-side door switch is on).

SPEEDOMETER CONTROL OUTLINE

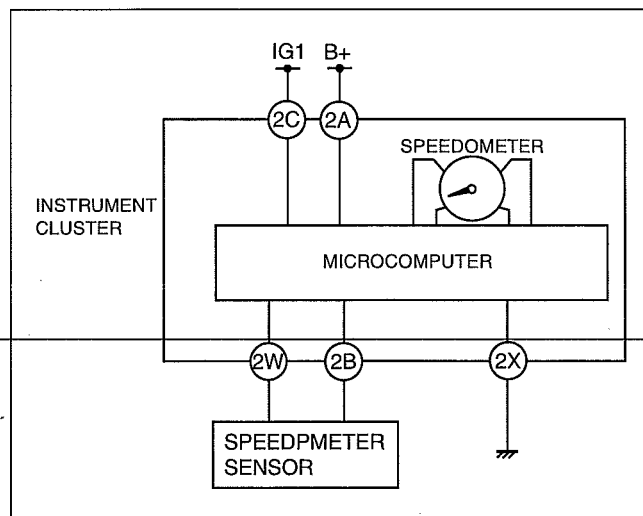
dcf092255430t15

- The vehicle speed signal is output from the speedometer sensor to the microcomputer in the instrument cluster.

SPEEDOMETER CONTROL CONSTRUCTION/OPERATION

dcf092255430t16

System Wiring Diagram



DCF922ZTB009

Operation

- The vehicle speed signal sent from the speedometer sensor is input to the microcomputer in the instrument cluster. The microcomputer calculates the current vehicle speed based on the vehicle speed signal, and sends an output signal to the speedometer.

INSTRUMENTATION/DRIVER INFO.

TACHOMETER CONTROL OUTLINE

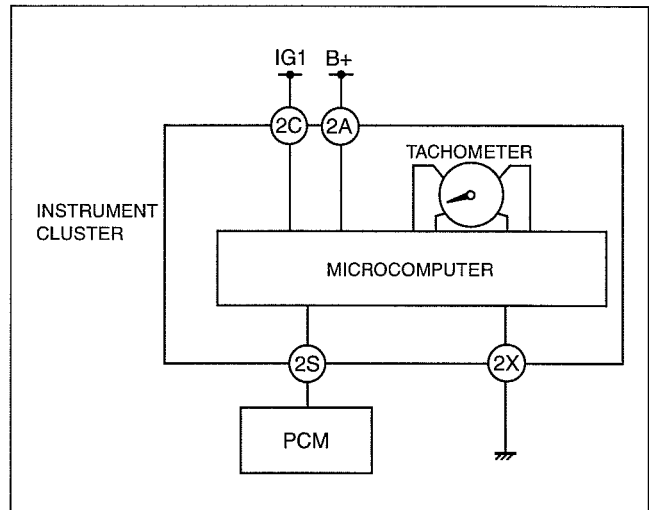
dcf092255430t17

- The engine speed signal is output from the PCM to the microcomputer in the instrument cluster.

TACHOMETER CONTROL CONSTRUCTION/OPERATION

dcf092255430t18

System Wiring Diagram



DCF922ZTBO10

Operation

- The engine speed signal sent from the PCM is input to the microcomputer in the instrument cluster. The microcomputer calculates the current engine speed based on the engine speed signal, and sends an output signal to the tachometer.

FUEL GAUGE CONTROL OUTLINE

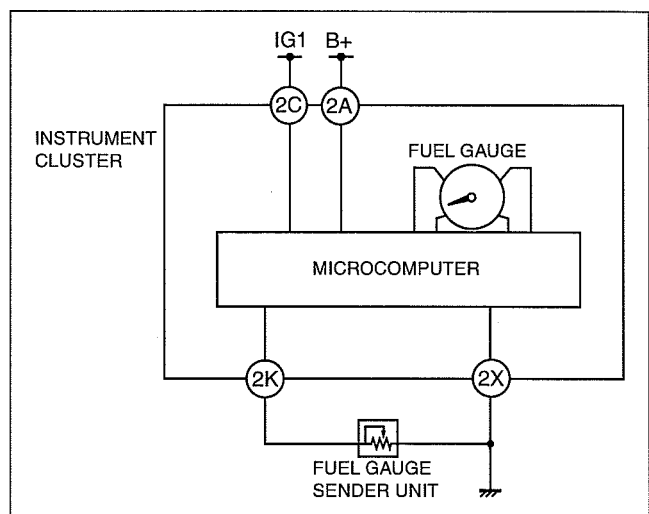
dcf092255430t19

- The fuel level signal is output from the fuel gauge sender unit to the microcomputer in the instrument cluster. Fuel gauge variation caused by fluctuating fuel level when cornering or driving on a slope, is reduced by microcomputer control.

FUEL GAUGE CONTROL CONSTRUCTION/OPERATION

dcf092255430t20

System Wiring Diagram



DCF922ZTBO11

Operation

- A resistance according to fuel level is sent from the fuel gauge sender unit to the microcomputer. The microcomputer calculates the average resistance within a specified time, and sends the output signal to the fuel gauge based on the calculated value.

INSTRUMENTATION/DRIVER INFO.

WATER TEMPERATURE GAUGE CONTROL OUTLINE

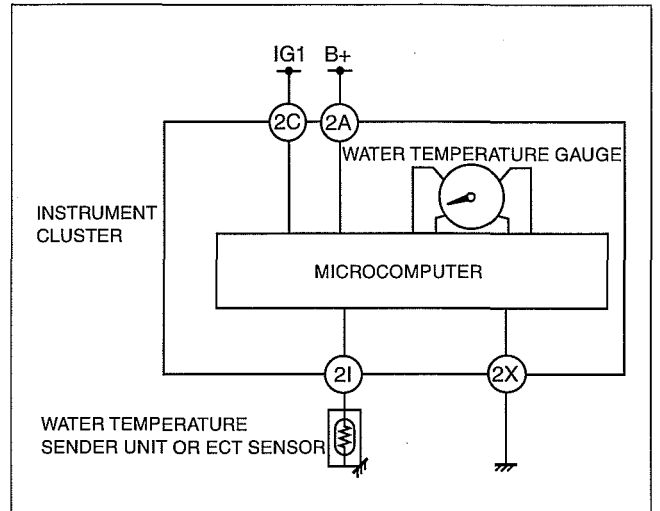
dcf092255430t21

- The engine coolant temperature signal is output from the water temperature sender unit or ECT sensor to the microcomputer in the instrument cluster.

WATER TEMPERATURE GAUGE CONTROL CONSTRUCTION/OPERATION

dcf092255430t22

System Wiring Diagram



DCF922ZTB012

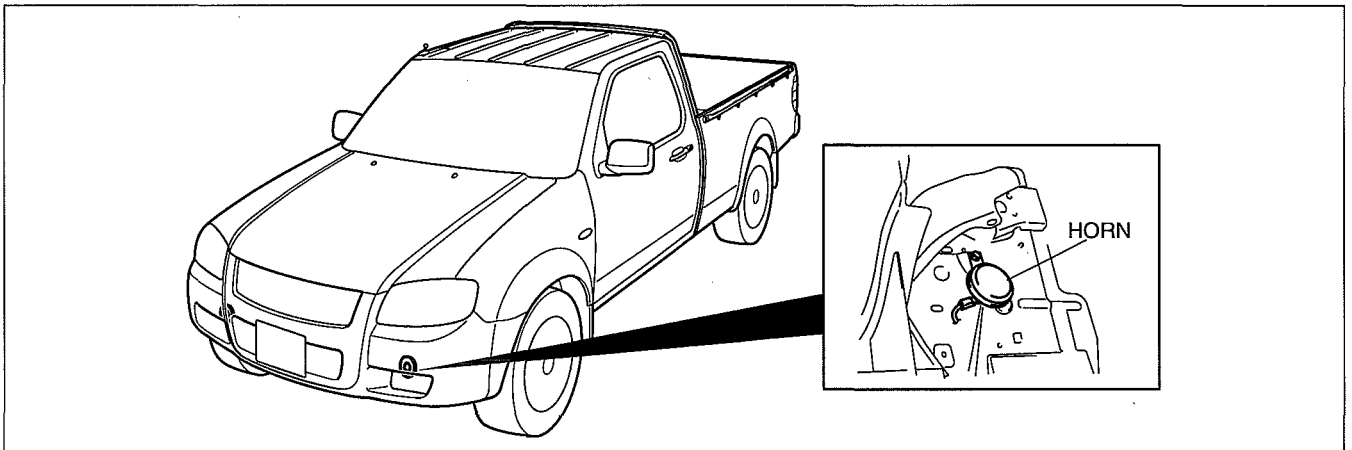
Operation

- The engine coolant temperature signal sent from the water temperature sender unit or ECT sensor is input to the microcomputer in the instrument cluster. The microcomputer calculates the current engine coolant temperature based on the engine coolant temperature signal, and sends an output signal to the water temperature gauge.

HORN CONSTRUCTION

dcf092266790t01

- A flat-type horn has been adopted.



DCF922ZTB013